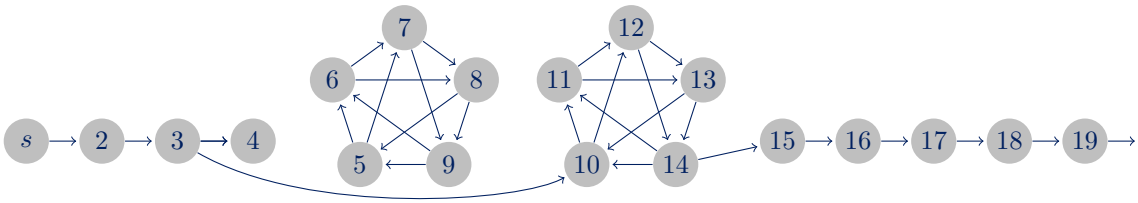
Kalman | Systems$^+$ Company Disclosure

Kalman | Systems$^+$



What We Do:

Quantum Computing: Advanced capabilities for optimisation, cryptography, and simulation | High-Performance Computing (HPC): Accelerating innovation through computational power | Data Science and AI: Predictive analytics, machine learning, and decision intelligence | Architecture Refactoring: Stabilising and modernising legacy systems for improved performance | Fractional Services: Cost-effective, flexible access to specialised expertise.

We want to help your entity access, future markets and what is available. Right now you will need to build life affirming technologies in a practical manner, without hallucinations. We do this by providing polymorphic access to our team of expertise, with fungible models and agile contracting and pricing, within reason and positivity[1].

[1] Practicality, prior to intellectual pursuit, is far superior

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1. Values

Honesty and integrity are cornerstones of Kalman Systems values, serving as guiding principles to ascertain clarity in confusion. In moments of uncertainty, we rely on our virtues to navigate rejection and plan for marriages and not divorce, solve problems; financially with resilience, mitigate risk, and multimodal loss. We are a group of technical and non technical professionals that understand endless consultancy and consistency is ***Tiring***. While we seek long term relationship with financial gain, good honest written and spoken communication, is preferred irrespective of stakeholders involved.

1.1 Ethics

NSPE Code of Ethics for Engineers[1] - Prescribed by the National Society of Professional Engineers for a corner stone of Kalman Systems(We, Me, I), priorities that infer indirectly, the *Hippocratic Oath* for engineering and science.

We hold paid memberships in the following organisations since 1998:

1. ASM International - Germany
   Head Office: 9639 Kinsman Road, Materials Park, OH 44073-0002, USA
   https://www.asminternational.org/
2. Association of Computer Machinery - USA
   Head Office: 1601 Broadway, 10th Floor, New York, NY 10019, USA
   https://www.acm.org/
3. Engineers Australia
   Head Office: 11 National Circuit, Barton ACT 2600, Australia
   https://www.engineersaustralia.org.au/
4. Institute of Electrical and Electronic Engineers - USA (IEEE)
   Head Office: 445 Hoes Lane, Piscataway, NJ 08854, USA
   https://www.ieee.org/
5. Institute of Engineering Technologists - UK (IET)
   Head Office: Michael Faraday House, Six Hills Way, Stevenage, Hertfordshire SG1 2AY, UK
   https://www.theiet.org/
6. American Mathematical Society - USA
   Head Office: 201 Charles Street, Providence, RI 02904-2294, USA
   https://www.ams.org/
7. SPIE – The International Society for Optics and Photonics
   Head Office: 1000 20th Street, Bellingham, WA 98225-6705, USA
   https://spie.org/

[1] https://www.nspe.org/resources/ethics/code-ethics

8. National Society of Professional Engineers
   Head Office: 1420 King St., Alexandria, VA 22314
   https://www.nspe.org/

1.2 Glossary

In formal communications and documents we encourage all to use the following Glossary terms as defined by: **ISO/IEC/IEEE 24765:2017 Systems and Software Engineering Vocabulary** [2]

1.2.1 Scope

"ISO/IEC/IEEE International Standard - Systems and software engineering-Vocabulary," in[3].

Consistent with ISO vocabulary standards, each technical committee is responsible for standard terminology within its area of specialisation. This glossary provides a common vocabulary applicable to all systems and software engineering work falling within the scope of ISO/IEC JTC 1/SC 7. The scope of each defined concept has been chosen to ensure a definition suitable for general application. In cases where a more restricted application is required, a more specific definition may be necessary.

The Glossary use consistent Software and Systems Engineering definitions, consistent with IEEE Computer Society Systems and Software Engineering Standards Committee (IEEE-CS S2ESC). Terms have been excluded if they were deemed prescriptive and has no context, to a particular group or organisation. The company proprietary or trademarked; multi-word terms whose meaning could be inferred from the definitions of their component words; or terms whose meaning within the field of information technology (IT) could be directly inferred from their common English dictionary definition.

Get your copy today!

[2] To avoid confusions as we can't offer a better Glossary ourselves.

[3] ISO/IEC/IEEE 24765:2017(E) , vol., no., pp.1-541, 28 Aug. 2017, doi: 10.1109/IEEESTD.2017.8016712. keywords: IEEE Standards; ISO Standards; IEC Standards; Information Technology; Software Engineering; Systems engineering and Theory; Terminology; Computer; dictionary; Information Technology; Software Engineering; Systems Engineering - 24765

2. Partners

Kalman Systems (We) perform business and market development while avoiding every sales technique known to the intelligent mind. We seek strategic global partnerships for innovation in strong economies with stable political environments, where permissible. Our focus is on business growth, leveraging commercially viable opportunities to deliver pertinent solutions and support. Delivered by qualified business referees, as an entity you are welcome to go directly to the partners of Kalman Systems, and you may refer to us or omit our involvement at your own discretion.

The engineering and scientific services are performed, provided, and delivered by Principal (Partner) Companies. Our role is to work in situ, ensuring alignment on deliverables, acting as a mediator when necessary, and initiating discussions on product vision with a focus on sustainable outcomes.

Commercial partners help us with functional and non functional considerations in work. These experts will give deliver any solution prescribed, after scope acceptance. New partnerships are always welcome, whether strategic or otherwise. We actively seek to engage with innovative minds, whether established or unrecognised.

While we do not seek to derive income from every transaction always, we prioritise economic growth and ease of business. However, as a company, we declare that most work will have financial imperatives, due to shared revenue and disgorgment of profits or services delivered. We are not precluded from receiving commissions on successful sales closures, where agreed upon with our partners, and disclosed upon request. Refer to further references to business services in chapter 8 provided mutually by our partners:

- aXlerate advisory – Australia [https://axelerateadvisory.com/] | Finance | Investments
- BDO Australia – Australia (Global) [https://www.bdo.com.au/en-au/home] | Global Operations, Taxation, Corporate Law
- CAN Automotion – Australia [https://www.canautomotion.com.au/] | Industrial | Mobile Machinery | Software and Automation
- Colakin – Australia [https://colakin.com] | Enterprise Architect | Software Engineering
- Compass Solutions – Australia | Mechanical | Engineering| Design | Fabrication
- ENPRO Envriotech – Australia [https://enproenirotech.com] | Renewable | Energy | System
- Ensemble Consulting Group – Australia [https://ensembletechnology.com/] | Management | Operations | Constraint Mangement
- Omnisure – Australia [https://omnisure.com.au/] | Commercial | Personal | Liability | Global

- Peak Capital – Australia [https://peakcapital.com.au/] | Finance | Investments
- Push Capital – Australia [https://pushcapital.com.au/] | Finance | Investments
- UCapital – Australia [https://www.ucapital.com.au/] | Finance | Investments
- NXON – India [https://nxon.io//] Electronics | Embedded | Software | Systems of Systems
- Tejasoft Innovations – India [https://tejasoft.com/] Fractional | Amortised | Enterprise | Systems | Technical Debt | Rumination |Services
- Tigon Technologies LTD DBA TLVTech – Israel | Middle East [https://www.tlvtech.io/] Software | Engineering | Concept | Development | Consultancy
- Annapurna Consulting – Indonesia | South East Asia | Hospitality | Consultancy | Mentoring
- Ergodic Solutions – South East Asia | Mainland China [https://ergodicsolutions.com/] Manufacturing | Large Scale | Operations Management | Research
- Bayshore Intelligence Solutions – USA | India [https://bayshoreintel.com/] Software | Engineering | Concept | Development | Consultancy
- QCI LLC – USA [https://quantumcomputinginc.com/] Quantum | Design | Fabrication | Semiconductors | Services
- VenturEsq LLC – USA (Global) | Finance | Investments | Global



3. *Capabilities* and Who are you?

Kalman Systems provides practical, high-value solutions with a flexible contracting model that emphasises agility, efficiency, and technology driven decision making. We specialise in providing solutions and support in advanced computing, engineering consulting, and business strategy services. We focus on many profitable domains, ensuring pragmatic engagement with one another. Half of our work relates to technical consulting and the remaining relates to business monetisation.

3.1 How to employ Kalman | Systems$^+$

- **Quantum Computing**: Offering optimisation, cryptographic solutions, and complex simulations to drive future-ready technology applications.
- **High-Performance Computing (HPC)**: Enabling businesses and research institutions to leverage computational power for accelerated innovation.
- **Data Science and Artificial Intelligence**: Utilising predictive analytics, machine learning, and decision intelligence to enhance operational efficiency and strategic planning.
- **Architecture Refactoring**: Modernising and stabilising legacy systems to improve performance, scalability, and security.
- **Fractional Services**: Providing organisations with flexible access to specialised expertise in a cost-effective manner.
- **Manufacturing Economy Consulting**: Supporting industrial firms in optimising processes, integrating emerging technologies, and achieving efficiency gains.
- **Platform Economy Consulting**: Assisting businesses in transitioning to digital platforms, optimising monetisation strategies, and managing technological shifts.
- **Business Management Consulting**: Advising on business modelling, risk assessment, operational strategies, and market entry approaches.

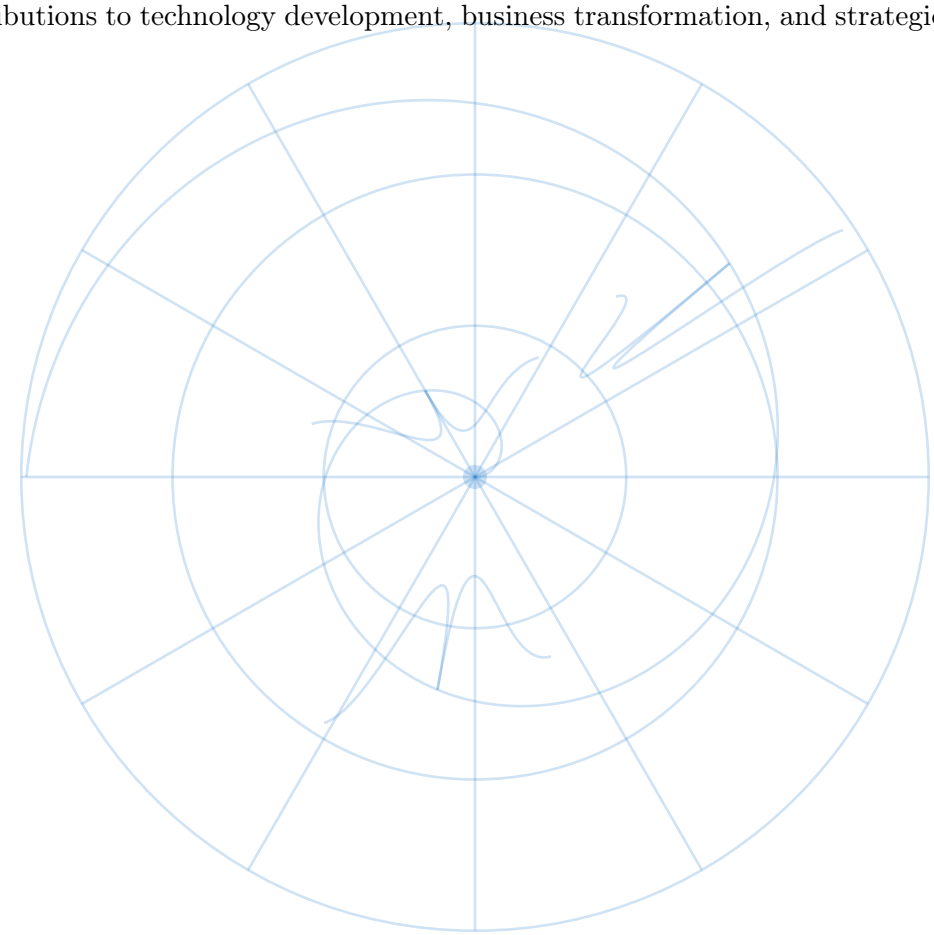
3.2 Audiences

At Kalman Systems, we are model-agnostic, ensuring a practical, tailored response to challenges. Our services span related and broader industries, and we acknowledge when we do not understand something. You may notice that we do not promote tools or frameworks in isolation, nor do we become unsettled by their use. Progenitors of solutions may emerge from irrelevant or irreverent beginnings; our aim is to provide you with a home, not to indoctrinate you on the selection of bricks.

- **Technology and Engineering Firms**: Seeking expertise in AI, high-performance computing, and system optimisation to drive innovation.

- **Industrial and Manufacturing Enterprises**: Companies aiming to enhance operational efficiency, integrate smart manufacturing solutions, and adopt Industry 4.0 principles.
- **Startups and SMEs**: Organisations requiring fractional CTO services, strategic business modelling, and growth-oriented consulting.
- **Government and Research Institutions**: Entities engaging in research and development of emerging technologies, looking for strategic insights and computational expertise.
- **Investment and Venture Capital Firms**: Investors interested in commercialisation strategies for deep-tech solutions, ensuring technological viability and market readiness.

By aligning our expertise with the needs of these stakeholders, Kalman Systems ensures impactful contributions to technology development, business transformation, and strategic market positioning.



4. Kalman | Systems^{+} Executive Summary

Kalman Systems was incorporated, Between two Directors, Mr. Mahesh Shastry. and Mrs. Praneetha Shastry, the company was incorporated in July 2022, NSW, Australia. With Guidance from BDO in Australia and specifically guided by discussion with Mr. Chris Balalovski and Mr. Simon Conolly on Governance, Taxation, and Relevant Laws.

With ethical agnostic business practices, leveraging mutual business enablement capabilities with engineering partners. We access authorised professional communities, processes, and tools necessary to implement measured model-driven approaches. Reduce technical debt and cruft[1].

In development classifications such as and not limited to all; Enterprise Software Systems | Systems Engineering | Electronics Hardware and Software Embedded Systems | Hybrid Systems | Complementary Co-Design Solutions | Industrial and Mechanical Design for Devices | Manufacture and Fabrication for Devices | Certification; we can delivery the complete development cycle till maintenance needs. The solution space is vast, and we focus on removing costly biases in requirements ensuring (-ity's) feasibility, flexibility, scalability, resilience and security in our propositions.

Kalman Systems celebrates industry capabilities with consulting and solutions across domains. Our expertise spans engineering, software development, business strategy, and cybersecurity, supporting organisations in optimising their operations and achieving sustainable growth. We assist in business and market development, enabling firms to expand strategically by identifying opportunities and mitigating risks. Our approach is based on fostering long-term partnerships, facilitating trade, and ensuring ease of business transactions. We support financial structuring, investment viability analysis, and operational resilience, particularly for technology-driven enterprises.

Our software and hardware development capabilities include systems integration, security assurance, and lifecycle management. We provide tailored solutions for high-performance computing, embedded systems, and digital baselines. Our expertise includes testing and assurance with reliability, compliance, and risk mitigation for mission-critical applications. In operations and supply chain management, we focus on process efficiency, manufacturing feasibility, and cost optimisation. We work with experienced partners to deliver end-to-end engineering solutions, including facility design and infrastructure planning.

Through data science, AI, and automation, we enhance decision-making processes, improve efficiency, and drive innovation. We also advise on technology roadmaps, cybersecurity frameworks, and regulatory compliance to ensure organisations remain resilient in a competitive landscape. We value collaborative engagement and seeks to establish relationships based on trust, expertise, and shared objectives. We offer practical, commercially viable solutions tailored to client needs, with a strong focus on sustainable growth and operational efficiency.

[1] Refers to unnecessary, redundant, or obsolete elements in a system, codebase | hardware principles, and design, accumulating over time. The term is commonly employed in software engineering, hardware design, and a general term in line with, technical contexts.

4.1 Agnostic Business Practices & Model-Driven Approaches

Kalman Systems operates with **agnostic business practices**, leveraging **mutual business enablement** with **engineering and technology partners**. We access **authorised professional communities**, established **processes**, and **industry-grade tools** to implement **measured, model-driven approaches** that reduce **technical debt and cruft**[2].

Our solutions address critical **engineering, software, and hardware development** challenges by eliminating **biases in requirements gathering**. Instead of relying on **assumptions or undefined specifications**, we focus on **quantifiable, testable, and adaptable design principles**. By ensuring **feasibility, flexibility, scalability, resilience, and security**, we create robust solutions that are **future-proof and adaptable to evolving business needs**.

4.1.1 Precision in Requirements Gathering

One of the key challenges in engineering and business solutions is **ambiguous or conflicting requirements**, which lead to **inefficiencies, misaligned expectations, and costly rework**. We use a product development model that is simple for you and brings technical contstraints in your reach, without undue elaboration.

4.2 Ensuring Clarity & Sustainable Growth

Kalman Systems is committed to **collaborative engagement**, establishing **long-term relationships based on trust and expertise**. We provide **practical, commercially viable solutions** tailored to each client's needs, ensuring **sustainable growth, operational efficiency, and continuous improvement**.

By focusing on **removing ambiguity in requirements gathering**, integrating **data-driven insights**, and applying **scalable engineering practices**, we enable businesses to **innovate, optimise, and remain competitive** in a rapidly evolving technological landscape.

4.3 Capabilities & Expertise

We collaborate with industry partners to provide **strategic consulting and engineered solutions across domains**, spanning, professional, engineering and emergent domains.

4.3.1 Professional Services Domain:

1. Consulting - Business | Strategy | Economic | Technology
2. Commercialisation
3. Business - Process | Modelling
4. Product Manufacturing | Industrial Design | Device Development
5. Engineering | Execution | Management
6. Quality | Assurance | Risk | Audit | Certification
7. Business Management Consulting
8. Operations Research and Management

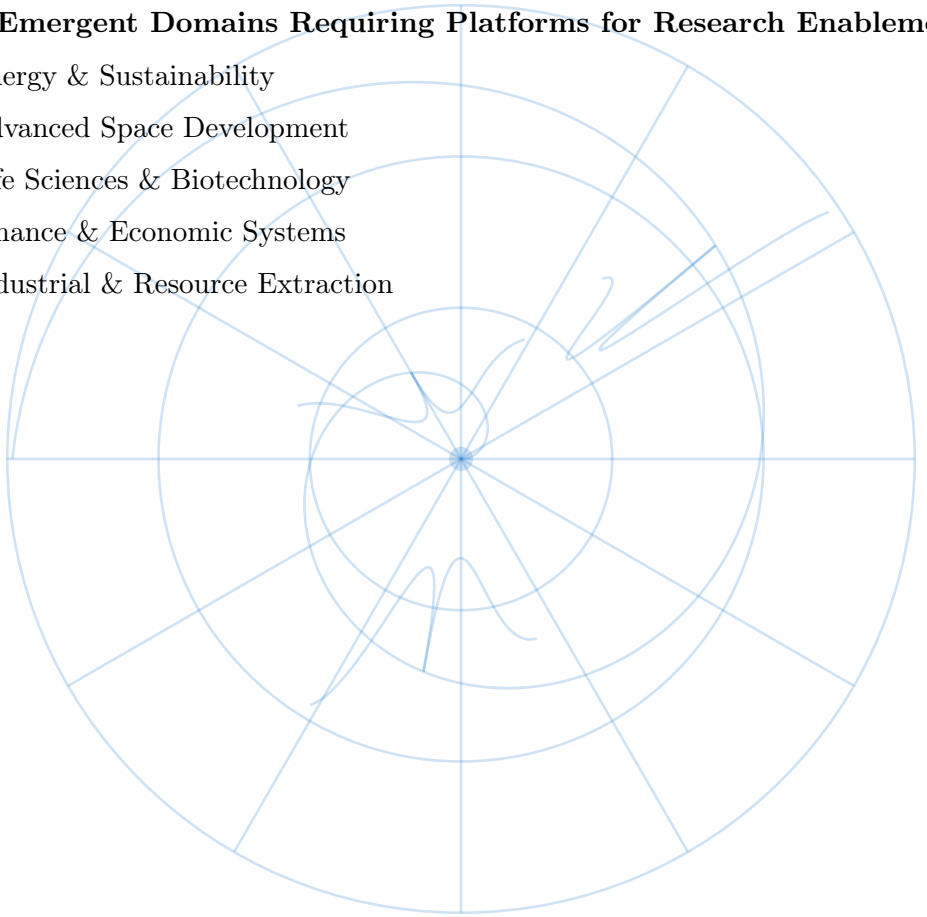
[2] Cruft refers to unnecessary, redundant, or obsolete elements in a system, codebase, or design that accumulate over time, impacting performance and maintainability.

4.3.2 Current Domains for Development:

1. Advanced Computing & Emerging Technologies
2. Cybersecurity & Decentralised Systems
3. Industrial & Manufacturing Technologies
4. Autonomous & Intelligent Systems
5. Extended Reality & Human-Centric Technologies
6. Pharmaceuticals, Healthcare & Medical Technologies
7. Web, Commerce & Service Ecosystems

4.3.3 Emergent Domains Requiring Platforms for Research Enablement:

1. Energy & Sustainability
2. Advanced Space Development
3. Life Sciences & Biotechnology
4. Finance & Economic Systems
5. Industrial & Resource Extraction



5. Modelling and Research Enablement

At Kalman Systems, with model-driven solutions[1] that help organisations navigate complexity with clarity and precision and system modelling, algorithmic development create stable low maintenance systems.

Our process begins with recognition and identification to define the system parameters. This foundational stage ensures that our models capture essential constraints, dynamics, and objectives, providing a strong basis for robust control. Next, in formulation, we can propose mathematical models and algorithmic frameworks to address system complexities. Using specialised solvers and numerical techniques, we construct adaptive models tailored to specific industry needs, from cybersecurity and high-performance computing to autonomous systems and financial systems.

We then move to solution mapping, ensuring that theoretical insights translate into practical, real-world applications. This phase involves simulation, empirical validation, and iterative testing, refining models to align with operational realities.Finally, we implement variance control, using statistical methods and machine learning to manage uncertainty and optimise model performance. This ensures stability, adaptability, and resilience against Ergodic[2] behaviour. For the formulation of Ergodic theorems please refer to Appendix A.

At Kalman Systems , we model to enable intelligent solutions that enhance efficiency, scalability, decision making expert systems. Whether addressing supply chain optimisation, cybersecurity architectures, or emerging technologies, our systematic methodology delivers measurable, results. Supported by theoretical rigour with real-world execution, we empower businesses, researchers, and policymakers to navigate an increasingly complex technological landscape with confidence.

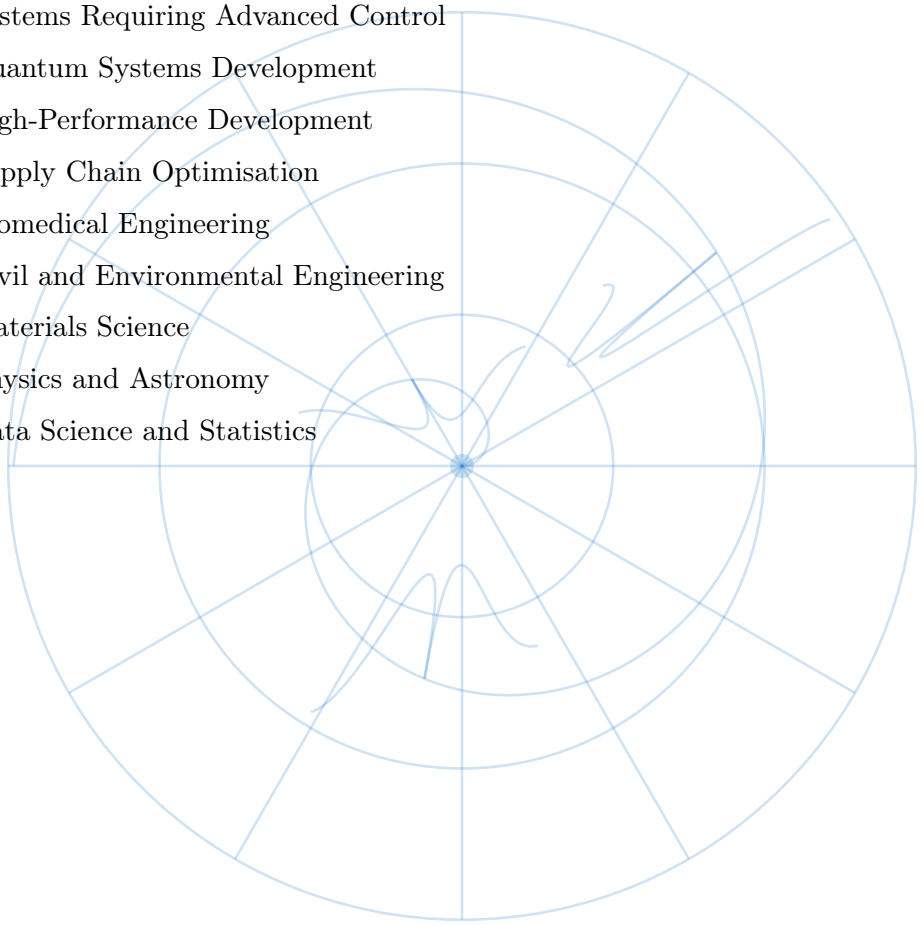
[1] *Hürilmann, D.T. (2024)* Mathematical Modeling Basics. 1st edn. Department of Informatics, CH-1700, Switzerland: University of Fribourg, pp. 28–34.

[2] https://en.wikipedia.org/wiki/Ergodicity

5.1 Disciplines in Engineering and Science

System Modelling requirements in modelling, statistics and experimental disciplines in engineering and science are fields where the development, application, and optimisation by algorithmic design & decomposition are critical for solving complex problems, analysing sensory[3], or designing systems. Below are some prominent modelling intensive domains we will serve:

1. Computer Science and Software Engineering
2. Electrical and Electronics Engineering
3. Mechanical and Aerospace Engineering
4. Autonomous Systems
5. Systems Requiring Advanced Control
6. Quantum Systems Development
7. High-Performance Development
8. Supply Chain Optimisation
9. Biomedical Engineering
10. Civil and Environmental Engineering
11. Materials Science
12. Physics and Astronomy
13. Data Science and Statistics



[3]Direct and Indirect.

6. Manufacturing Economy Consulting

Manufacturing Economy Consulting at Kalman Systems is designed to empower organisations in the manufacturing sector to navigate the complexities of modern industrial challenges while leveraging emerging opportunities. With a foundation in advanced technologies and process optimisation, we enable manufacturing firms to enhance productivity, reduce costs, and drive innovation. We have relationships with *Operations Management Consultants* that can support very large scale industries.

This consulting service integrates technology, strategy, and operations to achieve sustainable growth. By employing a reality driven approaches, we identify inefficiencies, optimise supply chains, and implement smart manufacturing solutions aligned with Industry 4.0 principles. Our expertise spans additive manufacturing and turnkey manufacturing systems in real-time.

Process Optimisation and Automation: We assist manufacturers in streamlining production workflows, reducing waste, and improving quality through AI-driven predictive maintenance, robotics, and IoT-enabled monitoring. Our strategies ensure agility and efficiency in dynamic markets.

Smart Manufacturing Solutions: We enable the adoption of smart factory initiatives by integrating sensors, machine learning, and advanced analytics into manufacturing ecosystems. These solutions provide real-time insights, enhancing adaptability and operational responsiveness.

Sustainable Practices and Green Manufacturing: Recognising the importance of sustainability, we help clients implement energy-efficient designs, integrate renewable energy, and adopt circular economy models to minimise costs and environmental impact.

Supply Chain Transformation: Leveraging blockchain and AI, we enhance supply chain transparency, reduce bottlenecks, and optimise logistics. Our experience in FMCG supply chains ensures efficient resource allocation and customer satisfaction.

Adoption of Emerging Technologies: We support manufacturers in integrating additive manufacturing, nanotechnology, and augmented reality (AR) for prototyping, assembly, and maintenance, ensuring competitiveness in rapidly evolving markets.

By aligning business objectives with technological advancements, Kalman Systems enables manufacturers to embrace innovation while maintaining operational resilience. Our tailored consulting services help clients meet the demands of modern manufacturing, ensuring success in a digital, global economy.

7. Platform Economy Consulting

Platform Economy Consulting at Kalman Systems focuses on enabling businesses to leverage digital platforms for growth, innovation, and market leadership. It helps organisations transition from traditional linear models to dynamic, ecosystem-driven structures where value is co-created with users and partners. By analysing trends, identifying opportunities, and optimising user engagement strategies, we guide clients in building scalable platforms.

Key areas include platform design, network effects, monetisation strategies, and governance models. Whether for marketplaces, subscription services, or community platforms, our consulting services empower businesses to unlock competitive advantages in an increasingly digitised economy. This approach fosters innovation, accelerates growth, and ensures sustained value creation.

In the Kalman Systems product strategy we create blueprints for creating and delivering value to customers, bespoke. Outlining the vision, objectives, and roadmap for a product, ensuring alignment with market demands and organisational goals. By identifying target audiences, understanding their needs, and analysing competing problem and solution pairs, product strategy provides clarity on positioning and differentiation. Key components include product lifecycle management, value propositions, and pricing strategies. A robust product strategy drives customer satisfaction, fosters loyalty, and positions the product as a market leader. Of course product monetisation only makes sense if the product is required by the public and in the public interest.

Our strategy provides an approach to leveraging business objectives for a competitive advantage. We assess current capabilities, identifying future needs and a roadmap for adoption, integration, and optimisation. Key areas include infrastructure planning, digital transformation, cybersecurity, and emerging technology exploration. A strong technology strategy enhances operational efficiency, fosters innovation, and improves adaptability in a rapidly evolving market.

7.1 Fractional or Holistic Service

Fractional services provide businesses with flexible, part-time access to expert professionals, including CTOs[1], and specialised technical professionals, while ensuring the provider retains allinged risks. This approach allows organisations to access impromptu expertise without the eternal commitment of FTE[2] Staff. Ideal for start-ups, SMEs, or companies in transition, fractional services offer tailored solutions to address specific challenges. By retaining risks, providers ensure accountability and reliability, enabling businesses to innovate confidently. This model supports growth, operational efficiency, and adaptability, empowering companies to focus on their core objectives while leveraging external expertise to navigate complex and dynamic environments.

Effective stakeholders supported by honest management is a cornerstone of Kalman Systems,

[1] Chief Technology Officer
[2] Full Time Equivalent

service delivery, ensuring that complex initiatives are executed efficiently and aligned with organisational goals.

7.1.1 Fractional SLDC and Operational Services(Everything OPS)

Fractional SLDC (Software Development Life Cycle) is a modular and flexible approach to managing the software development process. Unlike the traditional SLDC, where all stages are rigidly integrated and executed in a sequential or iterative manner, our fractional SLDC allows organisations to access and implement individual stages or subsets of the lifecycle based on their specific needs, resources, and goals. This approach is particularly useful for businesses with limited budgets, evolving requirements, or short-term project objectives. We can hire based on 6 fortnight sprints, long enough to remove technical debt, short enough to remain affordable.

7.1.2 Key Stages in Fractional SLDC

1. **Planning**: The process begins by identifying project goals, scope, stakeholders, and resource allocation. In a fractional model, planning can be tailored to address only immediate objectives or milestones.
2. **Requirements Analysis**: Focused efforts are made to gather and document functional and non-functional requirements, often on a feature-specific or phased basis.
3. **Design**: Architectural and system designs are created for specific components or modules instead of the entire system, aligning with the immediate development scope.
4. **Implementation**: Development is executed incrementally or modularly, delivering functional components that can be tested or integrated into existing systems. The implementation of planned sprints or fabrication of agreed design elements.
5. **Testing**: Testing efforts are localised to individual modules or specific functionalities, ensuring quality and compatibility before integration.
6. **Deployment**: Features or modules are deployed in phases, allowing for gradual implementation and user adaptation.
7. **Maintenance**: Ongoing support focuses on the deployed components, addressing issues, and implementing minor enhancements.
8. **Evaluation**: The process includes assessing outcomes to refine or improve future iterations or expansions.

7.1.3 Advantages of Fractional SLDC

- **Cost Efficiency**: Businesses only pay for the stages or services they require, reducing overheads.
- **Flexibility**: Adapts to changing needs, making it ideal for dynamic projects or startups.
- **Faster Delivery**: Focused efforts on specific stages accelerate time-to-market.
- **Risk Mitigation**: Gradual implementation and testing reduce risks allinged with large-scale failures.

Fractional SLDC empowers organisations to adopt a pragmatic and resource optimised constrained approach to software development, ensuring agility and alignment with business goals.

7.2 New Product Development:

Innovation is at the heart of Kalman Systems approach to new product development. The organisation identifies market needs and translates them into cutting-edge solutions that tackle emerging challenges. By leveraging deep expertise and a strong commitment to research and development, Kalman Systems ensures that its products are both innovative and future-ready.

Through light hearted detailed development practices, the organisation relies on advanced philosophies of work, rigorous testing, and industry best practices to create robust solutions tailored to your needs. By staying ahead of hype cycles and anticipating market shifts, Kalman Systems delivers products that not only meet current demands but also remain adaptable to future requirements.

Collaboration is central to this approach, with interdisciplinary teams working closely to refine ideas, prototype efficiently, and optimise performance. This commitment to excellence ensures that each product embodies reliability, scalability, and competitive advantage. By prioritising continuous improvement and strategic innovation, Kalman Systems empowers businesses to stay ahead in a rapidly evolving landscape.

7.3 Platform Integration and growth Services:

By employing a structured approach, Kalman Systems ensures that integration is smooth, minimising disruptions while maximising performance. Its expertise spans cloud computing, cybersecurity, AI, and enterprise systems, enabling clients to transition seamlessly without compromising existing operations. The organisation adopts best practices in system architecture, ensuring compatibility and future-proofing solutions for evolving business landscapes.

Collaboration plays a crucial role in this process, with teams working closely to understand operational challenges, streamline workflows, and optimise resource allocation. Kalman Systems leverages some advanced integration frameworks, automation, and achieving test coverage to guarantee seamless functionality. By focusing on adaptability and resilience, the organisation empowers businesses to operate more efficiently, reduce complexity, and enhance overall performance in a rapidly advancing digital world.

7.4 Platform and Systems Refactoring:

Refactoring is central to Kalman Systemss commitment to modernisation, enabling organisations to optimise existing system baselines and can develop realistic roadmaps, without losing face. By stabilising and enhancing architectures, Kalman Systems improves system constraints[3], improving reputation, cost management, and self esteem while maintaining operational continuity with security.

Through careful analysis, redundant or inefficient components are restructured, eliminating technical debt and enhancing maintainability. This approach reduces costs but also facilitates smoother system evolution, allowing businesses to adapt to emerging challenges with minimal staff complaints and resignations.

7.5 Testing

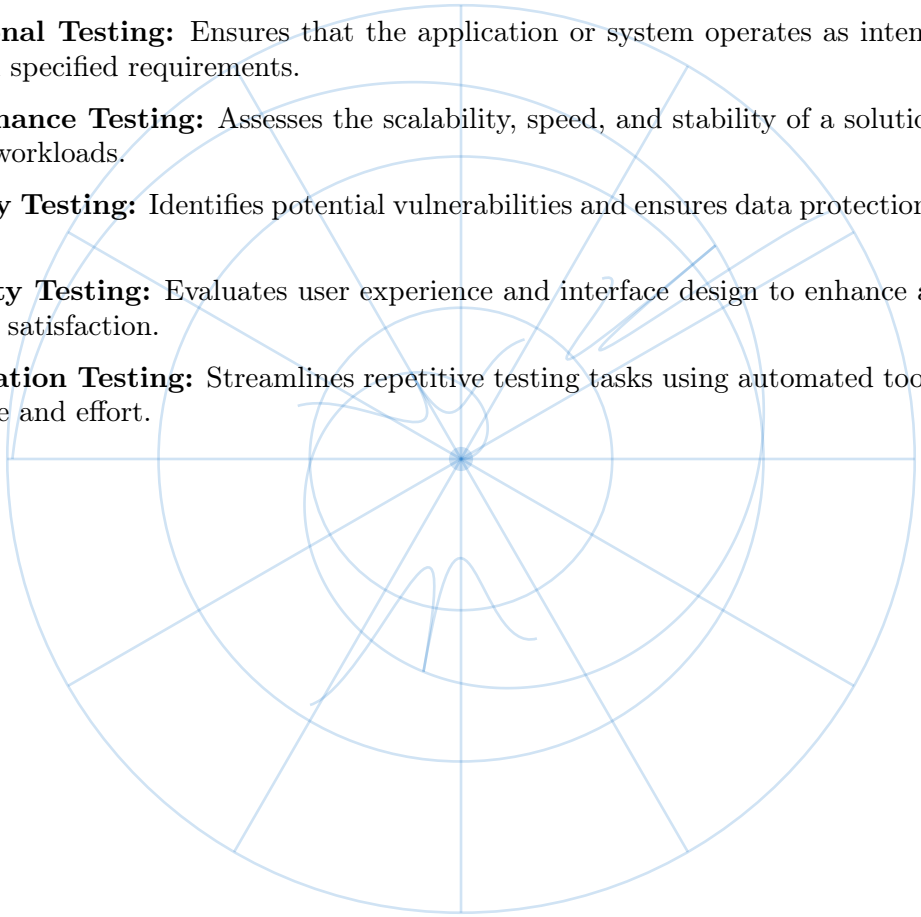
Testing services are essential for ensuring the quality, reliability, and performance of systems, applications, and products. At Kalman Systems, our testing framework is designed to identify and resolve defects, vulnerabilities, and inefficiencies before deployment, safeguarding operational integrity and user satisfaction.

[3]Theory of Constraints, https://www.tocinstitute.org

We employ a comprehensive approach that includes functional, performance, security, and usability testing to validate that solutions align with defined requirements and industry standards. Functional testing ensures core functionalities operate as intended, while performance testing evaluates system responsiveness and stability under various conditions. Security testing identifies potential threats and vulnerabilities, ensuring robust protection against cyber risks. Usability testing focuses on the end-user experience, guaranteeing intuitive and efficient interactions.

Our testing services play a critical role in risk mitigation, reducing long-term maintenance costs, and ensuring seamless integration across complex ecosystems. By adopting automated and manual testing methodologies, we enhance system resilience and accelerate time-to-market. Kalman Systems delivers rigorous testing solutions that support the deployment of high-performing, secure, and scalable technologies, reinforcing reliability, user trust and acceptance.

1. **Functional Testing:** Ensures that the application or system operates as intended and fulfils all specified requirements.
2. **Performance Testing:** Assesses the scalability, speed, and stability of a solution under various workloads.
3. **Security Testing:** Identifies potential vulnerabilities and ensures data protection against threats.
4. **Usability Testing:** Evaluates user experience and interface design to enhance accessibility and satisfaction.
5. **Automation Testing:** Streamlines repetitive testing tasks using automated tools, reducing time and effort.



8. Business Management Consulting

Business Management Consulting at Kalman Systems focuses on optimising organisational efficiency, strategy, and operations to drive sustainable growth.

Key areas include business strategy development, where we align organisational goals with market pain, and process optimisation on final requirements, ensuring operational safety through enterprise design, with patient persistence.

Kalman Systems Partners provide the following services:

- Acquisitions and Mergers
- Commercial Insurance
- Cross-Border Negotiations
- Empower Industry-Academic Collaborations
- Financial Planning | Banking Investments
- Intellectual Property and Legal Advocacy
- Operations Management Consultancy
- Provide Funding (Loans, Venture Capital)
- Provide Training
- Risk Management and Mitigation;

Commensurate to the scale of your disposition. Our expertise extends to change management, guiding businesses through transformations[1], with minimal effervescence.

Working with science & engineering verticals, and proven empirical techniques and industry standards with established practices, we enable market positioning, driving long term reproducible success.

8.1 Business Modelling

Business Modelling at Kalman Systems focuses on designing and analysing the mechanisms through which enterprises create, deliver, and capture value, even with no resources. It defines how an organisation operates, generates revenue, and sustains profitability by suturing frameworks and reality driven insights.

Key components include customer recognition, and presenting a realistic value propositions that meet market needs, and revenue stream identification, optimising financial sustainability. Cost structure analysis and key partnerships further enhance operational efficiency and strategic positioning.

[1] *Synchronised Crying with our Commiserations.*

Effective business modelling aligns an organisation with market complaints, enabling businesses to make informed decisions and adapt to evolving conditions. Whether launching new ventures or optimising existing operations, Kalman Systems provides tailored strategies that foster innovation, competitive differentiation, and a long-term peace of mind[2].

8.2 Business Process Modelling

Business Process Modelling (BPM) at Kalman Systems is a structured approach to understanding, analysing, and optimising organisational workflows. We clarify task execution, stakeholder roles, decision-making points, value chains in critical paths.

BPM helps identify inefficiencies, redundancies, and bottlenecks, enabling businesses to enhance productivity and operational agility. Through data and event driven knowledge and streamline resource allocation, reduce costs, and enhance service satisfaction.

By aligning business processes with strategic objectives, BPM ensures regulatory compliance, operational standardisation, and improved customer satisfaction. Legacy systems do not exist as long as they are required and not made imminently obsolete. The integration of incompatible services[3] or the expansion of capabilities through varied cross-functional collaboration is a core focus at Kalman Systems. We deliver tailored solutions that drive efficiency, scalability, and sustained competitive advantage.

8.3 Value and Consumer Modelling

Value and Consumer Modelling focuses on optimising the delivery of value to stakeholders while enabling businesses to scale efficiently based on real-time needs. By combining structured value analysis with flexible consumption models, organisations can enhance customer satisfaction, drive innovation, and achieve operational excellence.

Value Modelling involves identifying, defining, and quantifying the value generated through products, services, or processes. Ensuring investments align with business objectives enables organisations to differentiate offerings, enhance customer experiences, and maximise market impact by awareness of your intrinsic motivations, apart from money.

Consumer Business Modelling introduces a FMCG[4] paradigms that minimises upfront investments and optimises resource amortisation. Commonly applied in areas such as distributed computing, software and development services. The contemporary model provides flexibility while reducing operational complexities. Providers retain risks, ensuring reliability and accountability, while clients benefit from control of investment and outcomes.

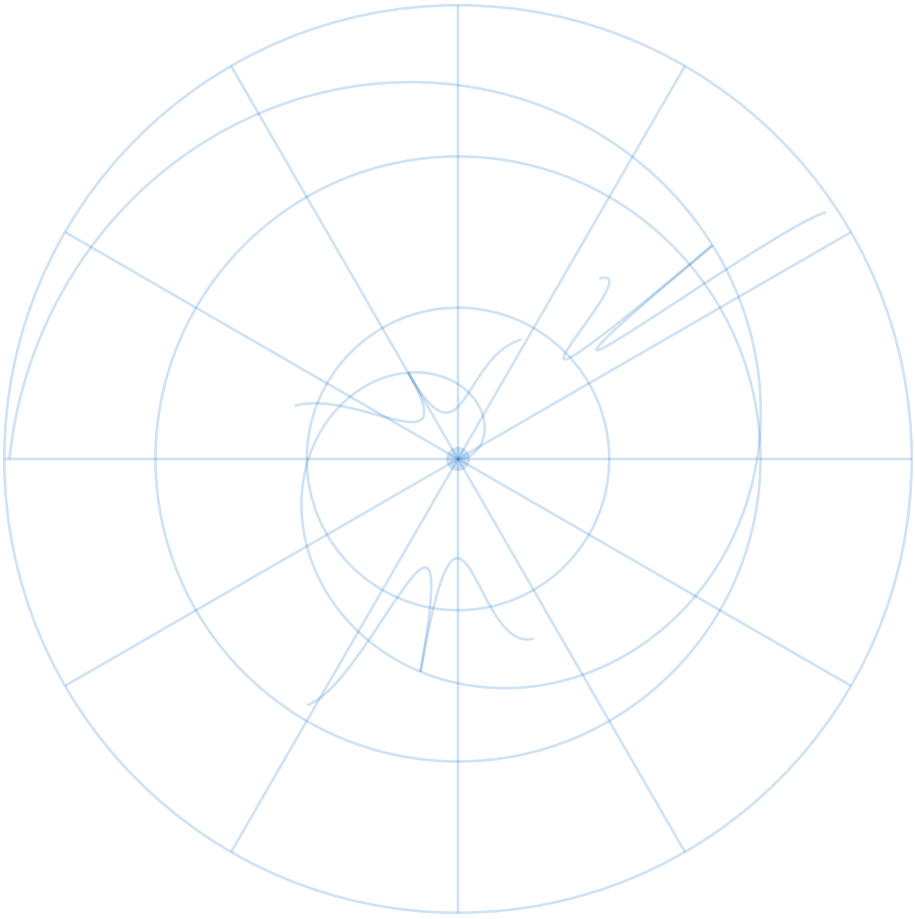
8.4 Audit Assurance Services

Audit Assurance Services at Kalman Systems establish a structured framework for assessing compliance, operational integrity, and financial accuracy. Through comprehensive evaluations of systems, processes, and controls, we ensure that organisations adhere to regulatory requirements, industry standards, and best practices.

Our audit methodology incorporates risk assessment, process evaluation, and strategic recommendations to identify inefficiencies, harmful biases and strengthen natural unforced governance. By providing actionable insights, we help businesses see clarity, responsible retention of care, and mitigate risks if not, reduce them.

[2] Prosperity is important, Never money, while time is definitely not Money.
[3] Systems.
[4] Fast Moving Consumer Goods.

Whether focusing on internal compliance, regulatory adherence, or operational audits, we deliver reliable and independent assurance. Our services enable organisations to build stakeholder trust, optimise operational efficiency, and ensure sustainable business success.



A. Ergodic Theorems

Deep in the vast reaches of space, the S.S. Smith charted its course through shifting cosmic tides. Onboard, its quantum AI; named Birkhoff; constantly monitored power levels, gravitational pulls, and sensor data, tirelessly logging each measurement into the ships' expansive databanks. Meanwhile, the ships' navigation intelligence; Von Neumann; streamlined all flight operations, adjusting thrust levels and plotting paths to minimise energetic waste.

Despite the ever-changing cosmic backdrop; radiation bursts, passing asteroid fields, and swirling nebulae; the S.S. Smith seemed to maintain a strangely consistent performance. Birkhoff and von Neumann attributed this reliability to Ergodic principles hardcoded into the ships' operational core. Every warp jump, every cruising segment, and every sensor reading fed into a grand cosmic puzzle that spanned both time and the ensemble of possible spacefaring conditions.

From Birkhoffs' perspective, the ships' subsystems were like tiny measure preserving transformations. Fuel consumption, energy flow, and mechanical wear were all carefully tracked so that each new second of flight was essentially a reconfigurable version of the same overarching system. Over long voyages, the time averages; like mean warp efficiency or average sensor fidelity; began converging on the ensemble averages that Birkhoff had calculated from the mountain of prior missions and simulations.

Occasionally, the S.S. Smith encountered an anomalous cosmic phenomenon; like a rogue black hole or an unexpected cluster of pulsars; that seemed to upend expected statistics. Yet, each time, Birkhoffs' built-in version of the Birkhoff Ergodic Theorem reassured the captain that, over the long haul, these fluctuations would be absorbed into the bigger picture. As long as the system remained measure-preserving; meaning total energy and momentum constraints were respected; the ultimate flight parameters would circle back to stable averages.

Simultaneously, von Neumanns' navigational AI seamlessly applied the Mean Ergodic Theorem to flight paths. By averaging each minute course correction over thousands of manoeuvres, von Neumanns' charts would converge on the single best route through any cosmic corridor. Short-term chaos; like asteroids swirling in unpredictable orbits; mattered less; the long-term trajectory fell into a predictable corridor of efficiency.

Over time, the crew came to trust these Ergodic laws as a kind of cosmic guarantee. The more they traversed uncharted space, the more data they collected, ensuring that the ships' predictive models grew stronger. From thermodynamic readings in the cargo bay to probabilistic analyses of signal reflection from distant quasars, each data point helped refine the ensemble. And as the voyage continued, each fresh moment at the helm felt less uncertain, guided by the knowledge that the S.S. Smiths' behaviour, though wildly varied in the short term, would always converge to a well-understood mean in the long term.

Thus, the Ergodic nature of the S.S. Smiths' operations served as the silent backbone of its cosmic expeditions. Birkhoff and von Neumann worked in tandem, weaving time averages and ensemble averages together into a single tapestry of predictive certainty. Whether traversing star-dense sectors or lonely galactic voids, the S.S. Smith sailed onward with unshakeable confidence; secure in the promise that long-term convergence was woven into its every circuit and

algorithm.

Let $(X, \mathcal{B}, \mu)$ be a probability space, and let $T : X \to X$ be a measure preserving transformation, meaning that for all $A \in \mathcal{B}$,

$$\mu(T^{-1}A) = \mu(A).$$

(1) Pointwise Ergodic Theorem (Birkhoff's Ergodic Theorem)

For any $f \in L^1(X, \mu)$, the time averages of f converge almost surely to the space average:

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \mathbb{E}[f|I](x),$$

where I is the σ-algebra of invariant sets under T, given by

$$I = \{A \in \mathcal{B} \mid T^{-1}A = A\}.$$

If T is Ergodic, meaning that every $A \in I$ has either $\mu(A) = 0$ or $\mu(A) = 1$, then the above reduces to

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \int_X f \, d\mu \quad \text{almost surely.}$$

(2) Mean Ergodic Theorem (von Neumann's Ergodic Theorem)

If T is a unitary operator on a Hilbert space H, then for any $f \in H$, the sequence of averages

$$P_N f = \frac{1}{N} \sum_{n=0}^{N-1} T^n f$$

converges in the norm of H to the orthogonal projection of f onto the subspace of T-invariant functions:

$$\lim_{N\to\infty} P_N f = Pf,$$

where P is the projection onto $\{g \in H \mid Tg = g\}$.

(3) Ergodic Decomposition

For any $f \in L^1(X, \mu)$, we have

$$\mathbb{E}[f|I](x) = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x),$$

which states that the Ergodic averages converge to the conditional expectation of f with respect to the invariant σ-algebra I. If T is Ergodic, this simplifies to

$$\mathbb{E}[f|I] = \int_X f \, d\mu.$$

B. Contact Us

$\pm \oint$ Mr. Mahesh Shastry.[1], $\prod$

Member of IEAust[2], IEEE[3], IET[4], ACM[5], ASM (Int.)[6], SPIE[7], AMS[8]

Director, Kalman | Systems$^+$

Inc. July 2022, NSW, Australia

ABN: 72 660 552 670 | ACN: 660 552 670

mshastry@acm.org

LinkedIn: https://linkedin.com/in/mshastryacmorg

Xing: https://www.xing.com/profile/Mahesh_Shastry

Meeting Booking Link: https://calendly.com/mshastryacmorg

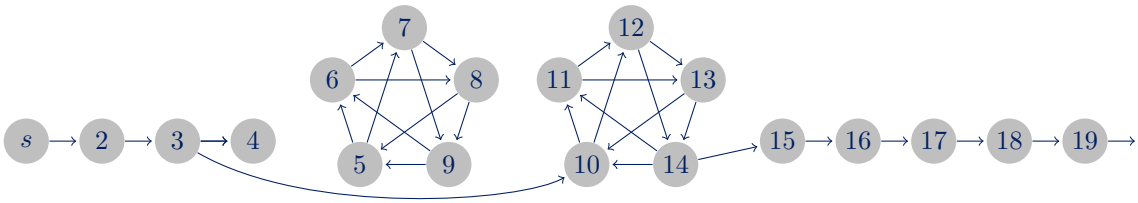
$\pm \oint$ Ms. Praneetha Shastry $\prod$

Non-Executive Director, Kalman Systems$^+$

ABN: 72 660 552 670 | ACN: 660 552 670

[1] *Ask me Why!*
[2] Engineers Australia
[3] Institution of Electrical and Electronic Engineers
[4] Institution of Engineering Technologists
[5] Association for Computing Machinery
[6] ASM International
[7] The international society for optics and photonics
[8] American Mathematical Society

Kalman | Systems^{+}



An Extensible :: Glossary | Bibliography ::

Kalman | Systems^{+} Annexe, composed of relevant glossaries, appendices; and reference bibliography collections; remains extensible and always growing. The document is written so as not to distract the audience from the central ideas of our business capability..

Kalman | Systems^{+} Annexe 1.0

Kalman | Systems+ Program Glossary

The glossaries contextualise consultancy, business, contracts, law, partnerships, ethics, conflict resolution, education, design, technology, and risk within the Kalman Systems Framework, ensuring clarity, accountability, and shared language that integrates technical, organisational, and ethical perspectives for resilient, adaptive, and trusted system delivery.

Consultancy

Consultancy within the Kalman Systems Framework reflects the structured engagement of expertise in complex environments. This glossary captures the terminology needed to design, deliver, and sustain advisory services. By clarifying terms such as due diligence, assignment, and client interface, it ensures that both consultants and clients can communicate in a disciplined way that supports repeatable quality and ethical delivery.

Business

The business glossary anchors Kalman Systems in the realities of markets, alliances, and functions. It defines the operational language that connects strategy with execution, from collaboration and competition to outsourcing and strategic alliances. Having a shared vocabulary of business terms underpins our ability to design value networks that are resilient, transparent, and oriented towards long-term sustainability.

Contract

Contracts embody the formalisation of relationships within Kalman Systems projects. This glossary articulates how obligations, remedies, and instruments are described and enforced. It provides a structured lens for understanding agreements, ensuring that both internal teams and external partners recognise how risks, responsibilities, and deliverables are apportioned. A common contractual language reduces ambiguity and supports trust.

Legal

The legal glossary reflects Kalman Systems commitment to regulatory alignment and ethical stewardship. Although not legal advice, it codifies the terms necessary to maintain internal clarity on compliance, governance, and obligations. By embedding legal terminology into our framework, we ensure that technical and business innovation are never divorced from accountability and the rule of law.

Partnerships

Partnerships and joint ventures represent collaborative models central to Kalman Systems scaling strategy. This glossary describes the fiduciary duties, shared responsibilities, and structural mechanisms that govern co-creation with external entities. It ensures that when we talk about alliances, we do so with precision, enabling equitable risk-sharing and value-creation across organisational boundaries.

Ethics

Ethics and governance provide the normative compass for Kalman Systems. This glossary introduces terms that shape how decisions are taken, how power is distributed, and how accountability is maintained. By foregrounding concepts such as stewardship, good faith, and ethical principles, we affirm that technical excellence must be guided by integrity, responsibility, and public trust.

Conflicts

Conflict, inevitable in human and organisational interaction, must be approached systematically. This glossary enumerates the methods of negotiation, mediation, conciliation, and arbitration that Kalman Systems recognises as part of its engagement toolkit. By making these processes explicit, we reinforce a culture that prioritises dialogue, fairness, and non-adversarial resolution wherever possible.

Education

The cognitive and learning glossary reflects Kalman Systems emphasis on knowledge as a living system. By defining terms related to research, cognition, and systems thinking, it provides a foundation for continuous improvement and organisational learning. This ensures that our framework is not static but adaptive, cultivating individual capability and collective intelligence.

Design

Mathematics within Kalman Systems is not abstract, but a working design tool. This glossary codifies the notation, objects, and modelling principles used across domains, ensuring that problem-solving has a rigorous shared base. From fundamental structures to applied models, it reflects the role of quantitative reasoning in shaping engineering, business analysis, and systemic innovation.

Technical

The technical glossary encapsulates the precise engineering vocabulary necessary for systems integration. Covering software, hardware, and standards such as ISO/IEC/IEEE 24765, it provides the controlled language that underpins reliability, interoperability, and safety. This ensures that Kalman Systems technical artefacts are designed, discussed, and delivered with consistency.

Risk

The risk glossary articulates how Kalman Systems perceives uncertainty, hazard, and control. By defining risk in terms of likelihood, consequence, and assurance, it enables disciplined management of exposure across projects. It integrates safety, resilience, and governance, ensuring that risk is neither ignored nor mystified, but measured, monitored, and mitigated in line with our framework.

Consultancy

A

Assignment

The transfer of contractual rights, property, or interest from one party (assignor) to another (assignee).

C

Collaboration Agreement

A contractual arrangement by which parties agree to cooperate in business or trade ventures, typically sharing risks and benefits.

Competition

The effort of two or more businesses to secure the same customers or market, subject to antitrust and fair-trade laws.

Consultant

An independent professional retained to provide expert advice or services in a specialised field, typically under a contract for services.

D

Due Diligence

The investigation and evaluation of the facts, risks, and obligations reasonably expected of a prudent businessperson before entering into a transaction.

F

Franchise

A contractual right granted by a franchisor to a franchisee to conduct business under the franchisors brand and system.

L

Licensing

The act of granting permission to use intellectual property, a brand, or technology under specified terms and conditions.

O

Outsourcing

The contracting out of a business function or process to an external provider, often for efficiency or cost savings.

S

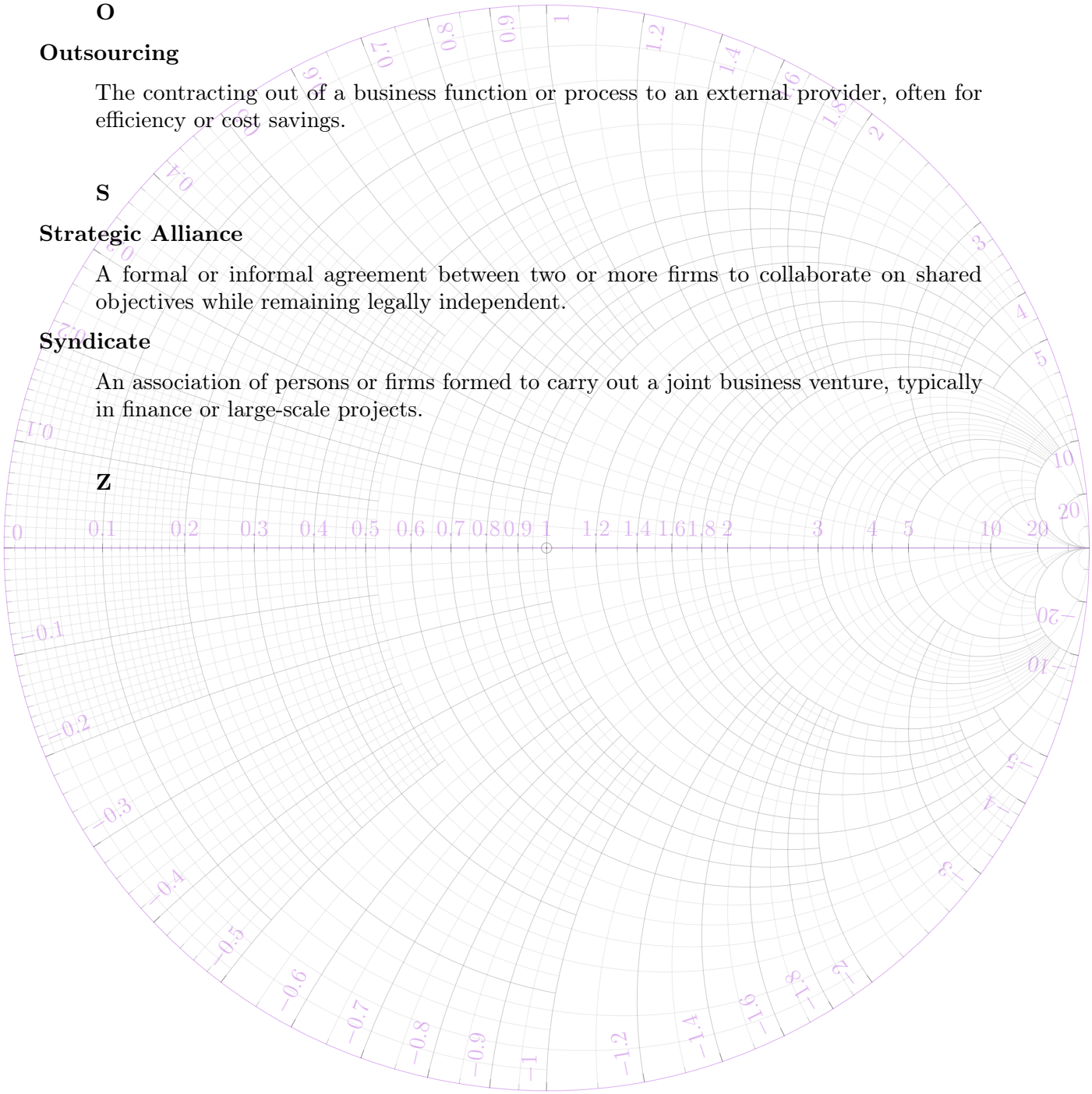
Strategic Alliance

A formal or informal agreement between two or more firms to collaborate on shared objectives while remaining legally independent.

Syndicate

An association of persons or firms formed to carry out a joint business venture, typically in finance or large-scale projects.

Z



Business

A

Agreement

A mutual understanding between two or more persons about their relative rights and duties regarding past or future performance (Black's Law Dictionary, 11th ed.)

C

Corporation

An entity created by law that is separate from its owners, having rights, privileges, and liabilities distinct from those of its members (Black's Law Dictionary, 11th ed.)

D

Disclosure

The act or process of making known something that was previously unknown or hidden; in business, often refers to financial or corporate transparency obligations (Black's Law Dictionary, 11th ed.)

G

Governance

The act or process of governing, especially corporate governance; the manner in which directors and officers of a company direct and manage its affairs (Black's Law Dictionary, 11th ed.)

L

Liability

The quality or state of being legally obligated or accountable; legal responsibility to another or to society (Black's Law Dictionary, 11th ed.)

M

Merger

The absorption of one organisation into another, such that only one entity remains after the transaction (Black's Law Dictionary, 11th ed.)

N

Negligence

Failure to exercise the standard of care that a reasonably prudent person would have exercised in a similar situation (Black's Law Dictionary, 11th ed.)

P

Partnership

A voluntary association of two or more persons who jointly own and carry on a business for profit (Black's Law Dictionary, 11th ed.)

Profit

The excess of revenues over expenditures in a business transaction; the gain after all costs are deducted (Black's Law Dictionary, 11th ed.)

R

Risk

The possibility of loss, injury, disadvantage, or destruction; in business, often linked to uncertainty in financial, legal, or operational contexts (Black's Law Dictionary, 11th ed.)

S

Shareholder

One who owns or holds a share or shares in a company, especially a corporation (Black's Law Dictionary, 11th ed.)

Statute

A law passed by a legislative body; in corporate practice, refers to statutes governing incorporation, fiduciary duties, and compliance (Black's Law Dictionary, 11th ed.)

T

Tort

A civil wrong, other than breach of contract, for which a remedy may be obtained, usually in the form of damages (Black's Law Dictionary, 11th ed.)

Z

Contracts

A

Acceptance

Manifestation of assent to the terms of an offer, creating a binding contract once communicated to the offeror (Blacks Law Dictionary, 11th ed.)

B

Bailment

Delivery of goods by the bailor to the bailee for a specific purpose, with obligation to return or otherwise dispose of them (Blacks, 11th ed.)

Bill of Exchange

A written, unconditional order by one party directing another to pay a certain sum of money to a third party; an essential negotiable instrument in trade (Blacks, 11th ed.)

Bill of Lading

Document of title issued by a carrier acknowledging receipt of goods for shipment; evidences the contract of carriage (Blacks, 11th ed.)

Breach of Contract

Failure to perform a contractual duty, whether material or minor, entitling the nonbreaching party to remedies (Blacks, 11th ed.)

C

Carriage of Goods

The legal regime governing contracts for transport of goods by land, sea, or air; includes carriers duties and shippers obligations (Blacks, 11th ed.)

Caveat Emptor

Latin for let the buyer beware; principle that buyer assumes risk in a sale unless protected by warranty or statute (Blacks, 11th ed.)

Charter Party

A contract by which a shipowner agrees to lease a vessel or space to a charterer for transport of goods (Blacks, 11th ed.)

Consignment

A commercial arrangement where goods are delivered to an agent (consignee) for sale on behalf of the owner (consignor) (Blacks, 11th ed.)

Contract

A promise or set of promises enforceable by law; breach gives rise to a remedy; requires offer, acceptance, and consideration (Blacks, 11th ed.)

Customs Duty

A tariff or tax imposed on goods imported or exported; administered by customs authorities (Blacks, 11th ed.)

D

Damages

Monetary compensation awarded for loss or injury caused by breach of contract or tort; includes compensatory and consequential damages (Blacks, 11th ed.)

Delivery

Voluntary transfer of possession from one party to another; essential to sales and mercantile transactions (Blacks, 11th ed.)

E

Ex Works (EXW)

A mercantile term where the seller makes goods available at their premises; buyer bears all costs and risks thereafter (Blacks, 11th ed.)

F

Force Majeure

A clause excusing nonperformance due to extraordinary events beyond control, such as war, strikes, or natural disasters (Blacks, 11th ed.)

G

Guarantee

A contractual undertaking by a guarantor to answer for the debt or obligation of another if that party defaults (Blacks, 11th ed.)

I

Incoterms

Internationally recognised commercial terms published by ICC defining allocation of costs and risks in trade; referenced in Blacks (11th ed.)

Indemnity

An obligation to make good anothers loss; often arises from contractual allocation of risk (Blacks, 11th ed.)

L

Letter of Credit (LC)

Banks undertaking to pay a beneficiary on presentation of conforming documents; essential in international trade (Blacks, 11th ed.)

Liquidated Damages

A contractual sum agreed in advance as compensation for breach, enforceable if a reasonable forecast of harm (Blacks, 11th ed.)

M

Mercantile Law

The body of commercial law governing merchants and trade, including sales, negotiable instruments, insurance, and partnerships (Blacks, 11th ed.)

Misrepresentation

False statement of material fact made to induce contract; renders agreement voidable and may support damages (Blacks, 11th ed.)

N

Novation

Replacement of one contract with another, or substitution of a new party, extinguishing the original obligation (Blacks, 11th ed.)

P

Payment

Performance of a contractual monetary obligation; in trade, may be made by cash, negotiable instrument, or credit transfer (Blacks, 11th ed.)

Pledge

A bailment of goods as security for a debt or obligation; creditor has lien until repayment (Blacks, 11th ed.)

Price

The consideration paid or promised in exchange for goods or services; essential element of sale (Blacks, 11th ed.)

R

Rescission

Cancellation of a contract, restoring parties to their pre-contractual positions, often for fraud or mistake (Blacks, 11th ed.)

S

Sale of Goods

A contract whereby the seller transfers or agrees to transfer property in goods to the buyer for a price (Blacks, 11th ed.)

Security Interest

A property interest created by agreement to secure payment or performance of an obligation; includes liens and charges (Blacks, 11th ed.)

Setoff

A debtors legal right to reduce the amount of a claim by any sum the creditor owes to the debtor (Blacks, 11th ed.)

Surety

One who promises to answer for anothers debt or duty; differs from guarantor by being directly liable (Blacks, 11th ed.)

T

Title

Legal right of ownership in property; in mercantile law, refers to passage of title in goods from seller to buyer (Blacks, 11th ed.)

W

Warranty

A contractual assurance of fact, quality, or performance of goods; breach allows for damages or replacement (Blacks, 11th ed.)

Z

Legal

A

Amendment

A contractual change agreed by the parties that alters terms or obligations; often documented as a written addendum (Blacks; 11th ed.)

Anti - Dumping Duty

A tariff imposed to counteract imports sold at less than fair value that injure domestic industry; traderemedy measure (Blacks; 11th ed.)

Arbitration

Binding dispute resolution before neutral arbitrator(s) pursuant to an agreement to arbitrate; award is enforceable subject to limited review (Blacks; 11th ed.)

B

Boycott

Collective or governmental refusal to deal with a person; entity; or country; sometimes subject to anti - boycott laws and sanctions regimes (Blacks; 11th ed.)

C

Choice of Law Clause

A contractual provision selecting the governing law for interpretation and enforcement of the agreement (Blacks; 11th ed.)

CIF

Cost; Insurance; and Freight; a sale term under which the seller arranges carriage and minimum insurance to the named port; risk allocation defined by trade usage and instruments like UCC or Incoterms; term appears in Blacks (Blacks; 11th ed.)

Confidentiality Agreement

A contract obligating parties to protect and limit use or disclosure of non - public information; also called an NDA (Blacks; 11th ed.)

Consideration

Bargained - for exchange supporting a promise; a legal detriment or benefit that makes a contract enforceable (Blacks; 11th ed.)

Consignee

The person to whom goods are shipped and who is named in the bill of lading for delivery (Blacks; 11th ed.)

Consignor

The person who delivers goods to a carrier for shipment; shipper of record (Blacks; 11th ed.)

D

Due Diligence

A measure of prudence or investigation; commonly the inquiry undertaken before transactions to assess legal; financial; and compliance risks (Blacks; 11th ed.)

E

Embargo

A governmental prohibition or restriction on trade with a foreign nation or on specific goods; often enforced through sanctions (Blacks; 11th ed.)

Entire Agreement Clause

A merger clause stating that the written contract represents the whole agreement; superseding prior negotiations (Blacks; 11th ed.)

F

Force Majeure

A contractual clause excusing performance when extraordinary events beyond a partys control prevent performance; e.g.; natural disasters; war; government action (Blacks; 11th ed.)

Forum Selection Clause

A contractual term designating the court or arbitral seat in which disputes must be brought; may be exclusive or nonexclusive (Blacks; 11th ed.)

Fraud

Knowing misrepresentation or concealment of material fact intended to induce reliance; causing damages; also covers deceit in commerce (Blacks; 11th ed.)

G

Governing Law

The body of law selected or applied to interpret a contract; resolve disputes; and determine remedies (Blacks; 11th ed.)

I

Injunction

A court order requiring a party to do or refrain from doing a particular act; used to prevent irreparable harm (Blacks; 11th ed.)

Insolvency

Financial condition in which liabilities exceed assets or debtor cannot pay debts as they come due; triggers bankruptcy or insolvency proceedings (Blacks; 11th ed.)

Intellectual Property

Intangible rights in creations of the mind; including patents; trademarks; copyrights; and trade secrets; frequently licensed in cross - border deals (Blacks; 11th ed.)

J

Jurisdiction

Power of a court to hear and decide a case; includes subject - matter and personal jurisdiction; significant in cross - border disputes (Blacks; 11th ed.)

L

Letter of Credit

A bank undertaking to pay the beneficiary upon compliant presentation of documents; facilitates international sales and allocates payment risk (Blacks; 11th ed.)

Limitation of Liability

A contractual cap or exclusion on damages or types of loss; subject to enforceability limits and public policy (Blacks; 11th ed.)

M

Most - Favored - Nation Treatment

A commitment to treat a party no less favorably than others; used in trade contexts and sometimes in commercial clauses (Blacks; 11th ed.)

P

Penalty Clause

A contractual sum intended to punish rather than compensate; generally unenforceable where characterised as a penalty rather than liquidated damages (Blacks; 11th ed.)

R

Representation; Warranty; Covenant

A representation states present or past fact; a warranty guarantees certain facts or quality; a covenant is a promise to act or refrain; each allocates risk and remedies (Blacks; 11th ed.)

S

Sanctions

Governmental measures; including trade; financial; and travel restrictions; used to influence or penalise states; entities; or persons (Blacks; 11th ed.)

Severability

A clause providing that if one provision is invalid; the remainder of the agreement continues in effect; preserving parties intent (Blacks; 11th ed.)

Specific Performance

An equitable remedy compelling performance of a contract; typically where goods or subject matter are unique; or damages are inadequate (Blacks; 11th ed.)

Subrogation

Substitution of one person in place of another with respect to a lawful claim; often insurers stepping into the insureds rights after paying a loss (Blacks; 11th ed.)

T

Tariff; Customs Duty

A schedule of duties imposed on imported or exported goods; assessed by customs authorities; includes ad valorem and specific duties (Blacks; 11th ed.)

Termination for Convenience; for Cause

Contractual end of performance either without breach under a reserved right; or for specified breach; subject to notice; cure; and wind - down provisions (Blacks; 11th ed.)

Title; Risk of Loss

Legal ownership and the allocation of risk for damage to goods; often defined by shipment terms and delivery documents; reflected in commercial law usage captured in Blacks (Blacks; 11th ed.)

Trade Secret

Information deriving independent economic value from not being generally known and subject to reasonable secrecy measures; misappropriation is actionable (Blacks; 11th ed.)

W

Waiver

Intentional relinquishment of a known right; may be express or implied by conduct; often constrained by no - waiver clauses (Blacks; 11th ed.)

Z

Partnerships

A

Affiliate

A corporation or entity that is related to another by shareholdings or control, such as a parent, subsidiary, or common control company (Blacks Law Dictionary, 11th ed.)

Association

An organised body of persons joined together for a common purpose, without necessarily forming a corporation or partnership (Blacks, 11th ed.)

C

Consortium

A group of companies or financial institutions that participate jointly in a large project or loan, sharing risks and benefits (Blacks, 11th ed.)

Cooperation Agreement

An agreement where parties coordinate efforts on specific business or consultancy projects, without creating a separate legal entity (Blacks, 11th ed.)

D

Distribution Agreement

A contract by which a manufacturer or supplier authorises a partner to market and resell its goods or services in specified territories (Blacks, 11th ed.)

F

Fiduciary Relationship

A relationship in which one party is bound to act for the benefit of another, such as partners, trustees, or corporate officers, with duties of loyalty and care (Blacks, 11th ed.)

I

Independent Contractor

A person or entity engaged to perform services while maintaining control over the manner and means of performance, not an employee or partner (Blacks, 11th ed.)

J

Joint Venture

An association of persons or entities combining property, knowledge, and efforts to carry out a single business enterprise for profit, with shared risks (Blacks, 11th ed.)

M

Memorandum of Understanding (MoU)

A nonbinding record of terms between parties who intend to formalise a future agreement, often used in cross-border consultancy (Blacks, 11th ed.)

S

Service Agreement

A contract under which a consultant, contractor, or partner agrees to deliver specific services, with obligations defined by scope, fees, and performance standards (Blacks, 11th ed.)

Shareholder Agreement

An agreement among a companys shareholders describing their rights, obligations, and management roles; used in joint ventures and consultancy entities (Blacks, 11th ed.)

Sponsorship Agreement

A partnership where one party supports another, financially or otherwise, in return for recognition or access to business opportunities (Blacks, 11th ed.)

Strategic Alliance

A cooperative arrangement between two or more independent firms to pursue agreed objectives while remaining separate legal entities (Blacks, 11th ed.)

T

Teaming Agreement

A pre-contract arrangement between companies to bid jointly on projects, setting out roles and responsibilities if the bid is successful (Blacks, 11th ed.)

Z

Ethics

A

Altruism

Unselfish regard for or devotion to the welfare of others; contrasted with egoism (Blacks Law Dictionary, 11th ed.)

C

Care Ethics

An ethical perspective emphasising duties of care, empathy, and maintaining relationships, often contrasted with rule-based morality

Communitarianism

A political and moral theory that emphasises the responsibilities of the individual to the community, and prioritises shared values

Conflict of Interest

A situation in which a persons duty to act in anothers interest is compromised by a competing personal or financial interest (Blacks, 11th ed.)

Confucian Ethics

A virtue-based philosophy stressing moral cultivation, harmony in social relations, and respect for roles and traditions

Contractualism

The ethical view that moral principles are those that free, rational individuals would agree to under fair conditions

Corporate Social Responsibility (CSR)

The concept that corporations have obligations beyond profit-making, including ethical duties to stakeholders, communities, and the environment

Cultural Relativism

The theory that moral judgments and norms are relative to cultural context and lack universal application

D

Divine Command Theory

The doctrine that what is morally right is what God commands, and what is wrong is what God forbids

E

Egoism

A theory holding that human conduct should be guided by self-interest; includes psychological and ethical egoism (Blacks, 11th ed.)

Ethical Pluralism

Recognition that more than one set of moral principles may be valid or applicable, depending on the context

Existentialist Ethics

Ethics emphasising freedom, authenticity, and responsibility of individuals in creating their own moral values

F

Fair Dealing

The duty to engage honestly and equitably in commercial transactions; often a standard in international trade law

Fiduciary Duty

The highest standard of care, requiring loyalty and good faith by fiduciaries such as directors or trustees toward beneficiaries (Blacks, 11th ed.)

H

Humanism

A philosophy prioritising human welfare, values, and dignity, grounded in reason rather than divine authority

K

Kantian Ethics

A deontological theory requiring actions to be based on duty and universal moral law rather than consequences

L

Libertarianism

A political philosophy prioritising individual liberty, free markets, and minimal governmental interference

M

Moral Particularism

The denial of absolute moral principles; judgments must depend entirely on context and circumstances

P

Pragmatic Ethics

A dynamic ethical approach assessing morality by its practical outcomes and adaptability to changing contexts

Professional Ethics

Standards governing conduct of members of a profession, emphasising integrity, competence, and avoidance of conflicts of interest

R

Research Integrity

Commitment to honesty, accuracy, transparency, and accountability in conducting and reporting scientific or academic research

Responsibility

A duty or obligation to act in accordance with law, morality, or contractual commitment; entails accountability for consequences (Blacks, 11th ed.)

Rule Utilitarianism

The doctrine that actions are right if they accord with rules whose general observance produces the greatest happiness

U

Utilitarianism

The doctrine that actions are right if they produce the greatest happiness for the greatest number

V

Virtue Ethics

An ethical system focused on the cultivation of virtuous character traits such as honesty, courage, and integrity

Z

Conflicts

A

Alternative Dispute Resolution (ADR)

A collective term for methods of resolving disputes outside courts, including negotiation, mediation, conciliation, and arbitration (Blacks, 11th ed.)

Arbitral Award

A final decision issued by an arbitral tribunal; may be subject to recognition or enforcement under treaties like the New York Convention (Blacks, 11th ed.)

Arbitration Agreement

A contract clause or separate agreement by which parties commit to resolve disputes by arbitration (Blacks, 11th ed.)

B

Bargaining

The process of discussion and compromise in order to reach terms of agreement (Blacks, 11th ed.)

C

Collective Negotiation

Negotiation between groups or representatives, common in labour and trade disputes, but also in joint ventures (Blacks, 11th ed.)

Conciliation

A nonbinding dispute-resolution process in which a neutral third party improves communication and proposes solutions, but parties decide outcome (Blacks, 11th ed.)

Counteroffer

A response to an offer that modifies terms, thereby rejecting the original offer and proposing new terms (Blacks, 11th ed.)

E

Enforcement of Award

The legal process by which an arbitral award or mediation agreement is recognised and executed in domestic or foreign courts (Blacks, 11th ed.)

F

Facilitation

A consensual process in which a neutral helps parties identify issues and explore solutions without imposing decisions (Blacks, 11th ed.)

G

Good Faith Negotiation

The duty to bargain honestly, fairly, and sincerely with the intent to reach agreement (Blacks, 11th ed.)

L

Letter of Intent (LOI)

A document recording parties preliminary agreement on major terms, often subject to negotiation and not binding unless expressly so (Blacks, 11th ed.)

Lien

A legal right or interest that a creditor has in anothers property, usually to secure payment of a debt (Blacks, 11th ed.)

M

Mediation

A method of dispute resolution involving a neutral third party who assists parties in reaching a voluntary, nonbinding agreement (Blacks, 11th ed.)

Mediation Settlement Agreement

A written record of parties resolution in mediation; enforceable in some jurisdictions by contract or statute (Blacks, 11th ed.)

R

Rescission

Cancellation of a contract and restoration of parties to pre-contract position, available for fraud, mistake, or misrepresentation (Blacks, 11th ed.)

S

Seat of Arbitration

The legal jurisdiction chosen as the procedural home of the arbitration, determining applicable law and court oversight (Blacks, 11th ed.)

Settlement Agreement

A negotiated resolution of a dispute; often includes releases, payments, or performance obligations (Blacks, 11th ed.)

Z

Education



A

Adaptability

The capacity to adjust effectively to changing conditions, environments, or expectations

Adversity

A state of hardship or difficulty that requires resilience and adaptive response in learning or decision-making

Agency

The capacity of an individual to act independently, make choices, and influence outcomes

Awareness

Conscious perception and understanding of internal states, external context, or social environment

B

Biological Stress

The physiological and biochemical response of an organism to internal or external demands that disturb equilibrium

Blooms Taxonomy

A hierarchical classification of cognitive learning objectives (remembering, understanding, applying, analysing, evaluating, creating); widely used in educational design

C

Cognitive Load

The total amount of mental effort being used in working memory; important in instructional design

Comfort

A psychological state of ease, safety, or confidence in a given environment or activity

Community

A group of individuals bound by shared goals, mutual learning, or collective practice

Context

The circumstances or background that give meaning to an action, concept, or statement

Courage

The willingness to confront fear, difficulty, or uncertainty in pursuit of goals or learning

Creativity

The ability to generate original and valuable ideas or solutions

Critical Thinking

Disciplined reasoning that evaluates evidence, arguments, and assumptions to reach justified conclusions

D

Discipline

A field of study or practice; also self-control and adherence to standards

E

Empathy

The ability to recognise, understand, and share the feelings and perspectives of others

Epistemology

The philosophical study of knowledge, including its nature, sources, and limits

Experience

Practical knowledge or skill gained through participation and observation

Experimentation

A systematic process of testing hypotheses or procedures to observe, validate, or discover knowledge

Exploration

The investigation of unfamiliar ideas or environments to gain knowledge or insights

F

Friendship

A trusted interpersonal relationship based on mutual respect, support, and affinity

G

Grammar

The structural rules of language governing syntax, morphology, and expression

H

Hermeneutics

The methodology of interpretation, especially of texts, symbols, or discourse

Heuristics

Simplified strategies or mental shortcuts used for decision-making and problem-solving

I

Innovation

The process of creating and applying new ideas, methods, or technologies

Intellectual Mobility

The ability to adapt, transfer, and apply knowledge flexibly across different contexts or disciplines

Intuition

Immediate understanding without conscious reasoning, often grounded in experience

L

Learning Curve

A graphical representation of how performance improves with experience or practice

Literature

Written or oral works valued for intellectual, cultural, or artistic significance

M

Meaning

The significance, purpose, or sense derived from experiences or symbols

Mentor

An experienced individual who provides guidance, feedback, and encouragement

Mentoring

The interactive process of support and guidance between a mentor and mentee

Mentorship

The formal or informal relationship in which mentoring occurs

Metacognition

Awareness and regulation of ones own thinking and learning processes

O

Ontology

The study or classification of entities and their relationships within a system of knowledge

P

Pathways

A structured educational programme offering sequential learning experiences

Philosophy

The systematic study of fundamental questions regarding existence, values, reason, and language

Prescience

Foresight; the ability to anticipate or predict future events

Presence

The state of being engaged and attentive, projecting awareness and confidence

Problem-Solving

The process of identifying, analysing, and resolving issues through reasoning or experimentation

Programme

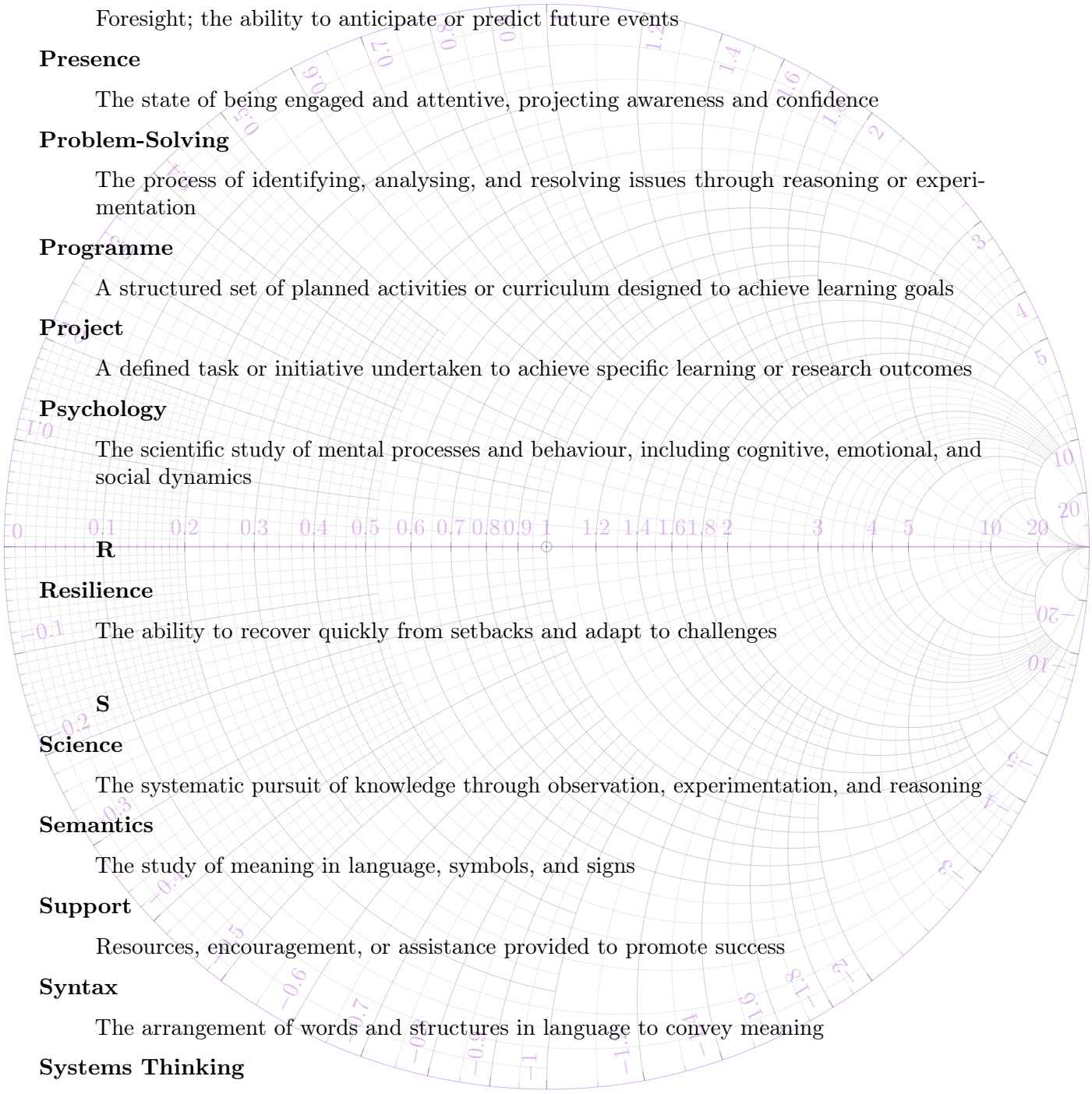
A structured set of planned activities or curriculum designed to achieve learning goals

Project

A defined task or initiative undertaken to achieve specific learning or research outcomes

Psychology

The scientific study of mental processes and behaviour, including cognitive, emotional, and social dynamics



R

Resilience

The ability to recover quickly from setbacks and adapt to challenges

S

Science

The systematic pursuit of knowledge through observation, experimentation, and reasoning

Semantics

The study of meaning in language, symbols, and signs

Support

Resources, encouragement, or assistance provided to promote success

Syntax

The arrangement of words and structures in language to convey meaning

Systems Thinking

A holistic approach to problem-solving that views issues as parts of an interconnected system

T

Taxonomy

A systematic classification framework, often hierarchical, for organising concepts

Tests

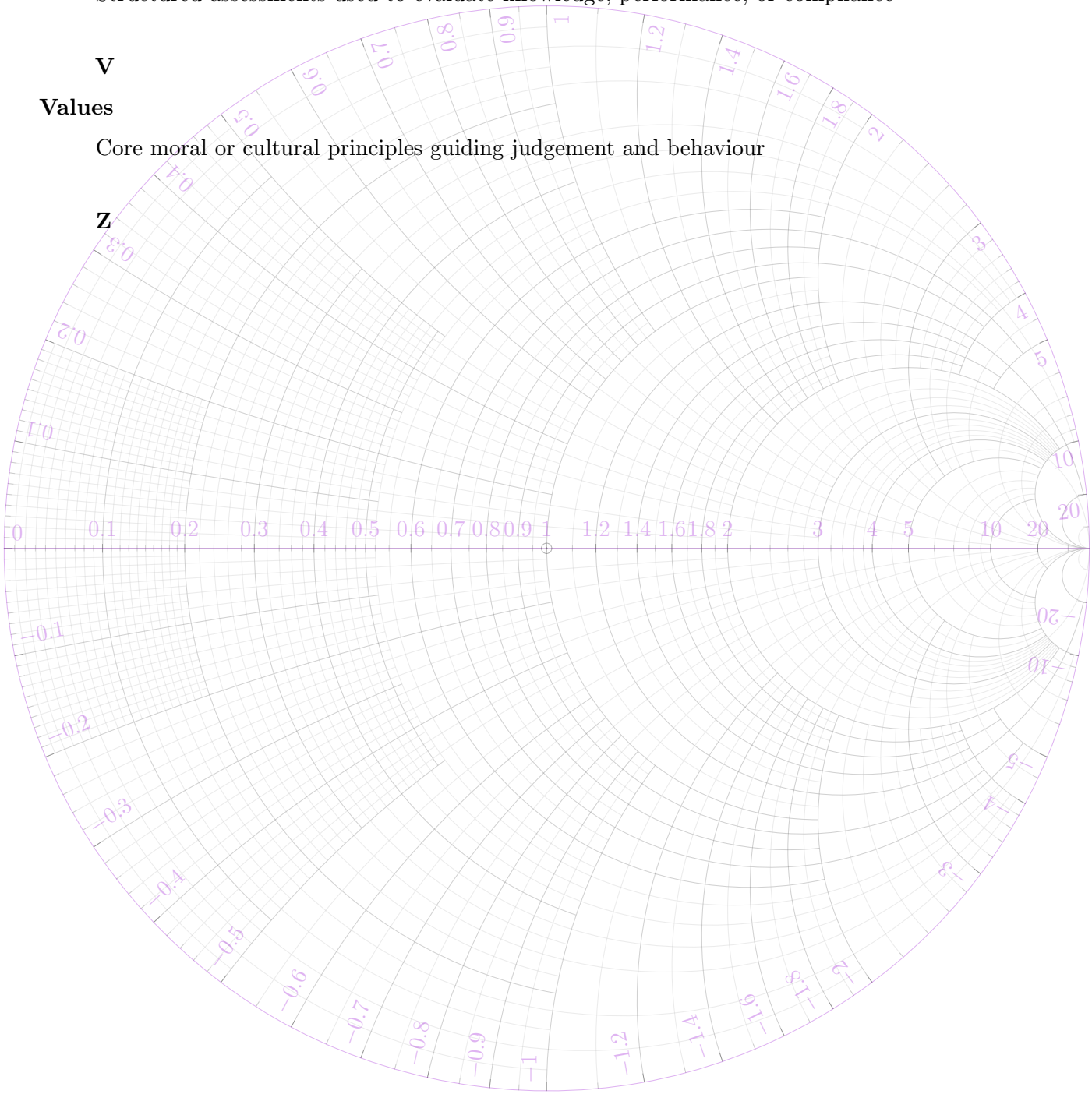
Structured assessments used to evaluate knowledge, performance, or compliance

V

Values

Core moral or cultural principles guiding judgement and behaviour

Z



Design

glssymbols

L^p Space

Equivalence classes of measurable functions with finite $p$norm; Banach for $1 \le p \le \infty$

σAlgebra

A collection closed under complements and countable unions; supports measurable sets

A

Algebraic Structure

A set equipped with operations satisfying axioms; e.g. groups; rings; fields; modules

Analytic Function

Complex differentiable on a domain; power series expansion; rigidity phenomena

B

Banach Space

A complete normed vector space; supports HahnBanach; open mapping; and duality

Bifurcation

Qualitative change in solution structure as a parameter varies; folds; Hopf; period doubling

Bounded Operator

A continuous linear map between normed spaces; has finite operator norm

Brownian Motion

Continuous time Gaussian Markov process with independent increments; It integral foundation

C

Calculus of Variations

Extrema of functionals; EulerLagrange equations; basis for weak formulations

Category; Functor

Objects and morphisms with composition; functors preserve structure between categories

Cauchy Integral Formula

Values of an analytic function determined by boundary integral; yields estimates

Central Limit Theorem

Properly normalised sums converge in distribution to a normal law under broad conditions

Chaos

Sensitive dependence; topological mixing; dense periodic points; typical in nonlinear maps

Compact Operator

Maps bounded sets to relatively compact sets; discrete spectra in many settings

Compactness

Every open cover admits a finite subcover; yields extreme value and convergence results

Conditioning; Stability

Sensitivity of solutions to data; stability of algorithms under perturbations

Connectedness

A space not decomposable into two disjoint nonempty open sets; underlies path behaviour

Consistency; Convergence

Discretisations approximate continuous models; Lax equivalence for linear well posed problems

Convex Function

Epigraph is convex; subgradients; Jensens inequality; dual representations

D

Differentiable Manifold

A Hausdorff; second countable space locally like R^n with smooth transition maps

Distribution; Generalised Function

Continuous linear functional on test functions; extends differentiation and PDE solutions

Dominated Convergence Theorem

Interchanges limit and integral under domination by an integrable function

Duality

Primaldual pairs; weak and strong duality; saddle points; certificates of optimality

Dynamic Programming

Optimal substructure; Bellman equations; value iteration in sequential decision problems

E

Eigenvalue; Eigenvector

$Ax{=}\lambda x$; spectral quantities governing stability; modes; and response

Entropy

Shannon measure of uncertainty; links to coding; inference; and statistical mechanics

Equivalence Relation

A relation that is reflexive; symmetric; transitive; partitions a set into equivalence classes

Ergodic Theorem

Time averages equal space averages under ergodicity; links dynamics and probability

Expectation; Variance

Mean and dispersion of a random variable; central to risk and uncertainty

F

Filtration; Stopping Time

Growing information sets; stopping times measurable with respect to filtrations

Finite Difference Method

Grid based approximations of derivatives; order; stability; CFL type conditions

Finite Element Method

Variational discretisation using basis on meshes; error estimates; adaptivity

Fourier Transform

Integral transform mapping convolution to multiplication; central in analysis and PDE

FubiniTonelli Theorems

Criteria for exchanging order of integration and summation on product spaces

Function

A rule assigning to each element of a domain exactly one element of a codomain

G

Game Theory; Nash Equilibrium

Strategic interaction; equilibria where no player gains by unilateral deviation

Geodesic

Curve locally minimising length with respect to a Riemannian metric; satisfies geodesic equation

Graph

Vertices and edges modelling pairwise relations; degree; paths; cycles; connectivity

Group

A set with associative operation; identity; inverses; symmetry and invariants

H

HahnBanach Theorem

Extends bounded linear functionals; yields separation and duality principles

Hausdorff Space

Any two distinct points have disjoint neighbourhoods; ensures separated limits

Hilbert Space

A complete inner product space; geometry via orthogonality; projections; spectral theorems

I

Iterative Solvers

Krylov subspace methods; preconditioning; multigrid for large sparse systems

It Calculus

Integration and stochastic differential equations with respect to semimartingales

K

KKT Conditions

First order optimality for constrained convex programmes; generalises Lagrange multipliers

KullbackLeibler Divergence

Asymmetric measure of separation between probability laws; convexity and variational forms

L

Lagrange Multipliers

Stationarity with constraints via Lagrangian; primal feasibility; complementary slackness

Law of Large Numbers

Sample averages converge to the mean under mild assumptions; weak and strong forms

Lebesgue Integral

Integral defined via simple functions and measure; enables convergence theorems

Lie Group; Lie Algebra

Smooth group with compatible manifold structure; algebra from left invariant vector fields

Linear Operator

A linear map between vector spaces; often represented by a matrix in finite dimensions

Lyapunov Stability

Qualitative stability via Lyapunov functions; asymptotic and exponential variants

M

Markov Process

Future depends on present state only; transition semigroups; generators; equilibrium

Martingale

Adapted process with constant conditional expectation; optional stopping; inequalities

MaxFlow MinCut

Equality of maximum flow value and minimal cut capacity in networks; duality manifestation

Measurable Function

A function whose preimages of Borel sets are measurable; integrates via Lebesgue theory

Measure

A countably additive set function; generalises length; area; probability

Metric Space

A set with a distance satisfying positivity; symmetry; triangle inequality

Module

Generalises vector spaces by allowing scalars from a ring; central to linear representations

Monte Carlo Method

Random sampling for numerical integration and simulation; variance reduction schemes

N

Natural Transformation

A morphism between functors; expresses canonical comparison of constructions

O

Ordinary Differential Equation

Equation involving derivatives with respect to one variable; initial value problems

P

Partial Differential Equation

Relations among partial derivatives of multivariate functions; elliptic; parabolic; hyperbolic

Partial Order

A reflexive; antisymmetric; transitive relation; defines posets and lattices

Plancherel Theorem

Isometry of L^2 under Fourier transform; energy preserved between domains

Probability Space

(Ω, F, P) with sample space; σalgebra; probability measure

R

Random Variable

Measurable map from Ω to a measurable space; induces a distribution

Relation

A subset of a Cartesian product; encoding pairwise associations such as order or equivalence

Residue

Coefficient of $1/(z - z_0)$ in Laurent expansion; computes contour integrals

Riemannian Metric

Smooth inner product on tangent spaces; geodesics; curvature; comparison theorems

Riesz Representation

Identifies duals of certain function spaces; e.g. $H \cong H^*$ via inner product

Ring; Field

A ring has two operations with distributivity; a field has multiplicative inverses for nonzero elements

S

Set

A well specified collection of objects; the basic carrier for mathsematical structures

Singular Value Decomposition

Factorisation $A = U\Sigma V^*$; *reveals rank; conditioning; principal directions*

Sobolev Space

Function space with weak derivatives in L^p; controls regularity of PDE solutions

Spanning Tree

A cycle free connected subgraph containing all vertices; minimal connectors

Spectral Radius

$\rho(A) = \max |\lambda|$; *key to convergence of iterations and stability of dynamical schemes*

Spectral Theorem

Diagonalisation of normal operators on Hilbert spaces via spectral measures

State Space Control

Linear systems $\dot{x} = Ax + Bu$; controllability; observability; feedback design

Subgradient; Proximal Map

Generalised gradients for nonsmooth convex functions; proximal operators for splitting

T

Tangent Space; Vector Field

Linearisation of a manifold at a point; derivations; flows of vector fields

Tensor; Tensor Product

Multilinear objects and their universal construction; encodes coupling of spaces

Topological Space

A set with a topology of open sets; capturing continuity abstractly

V

Vector Space

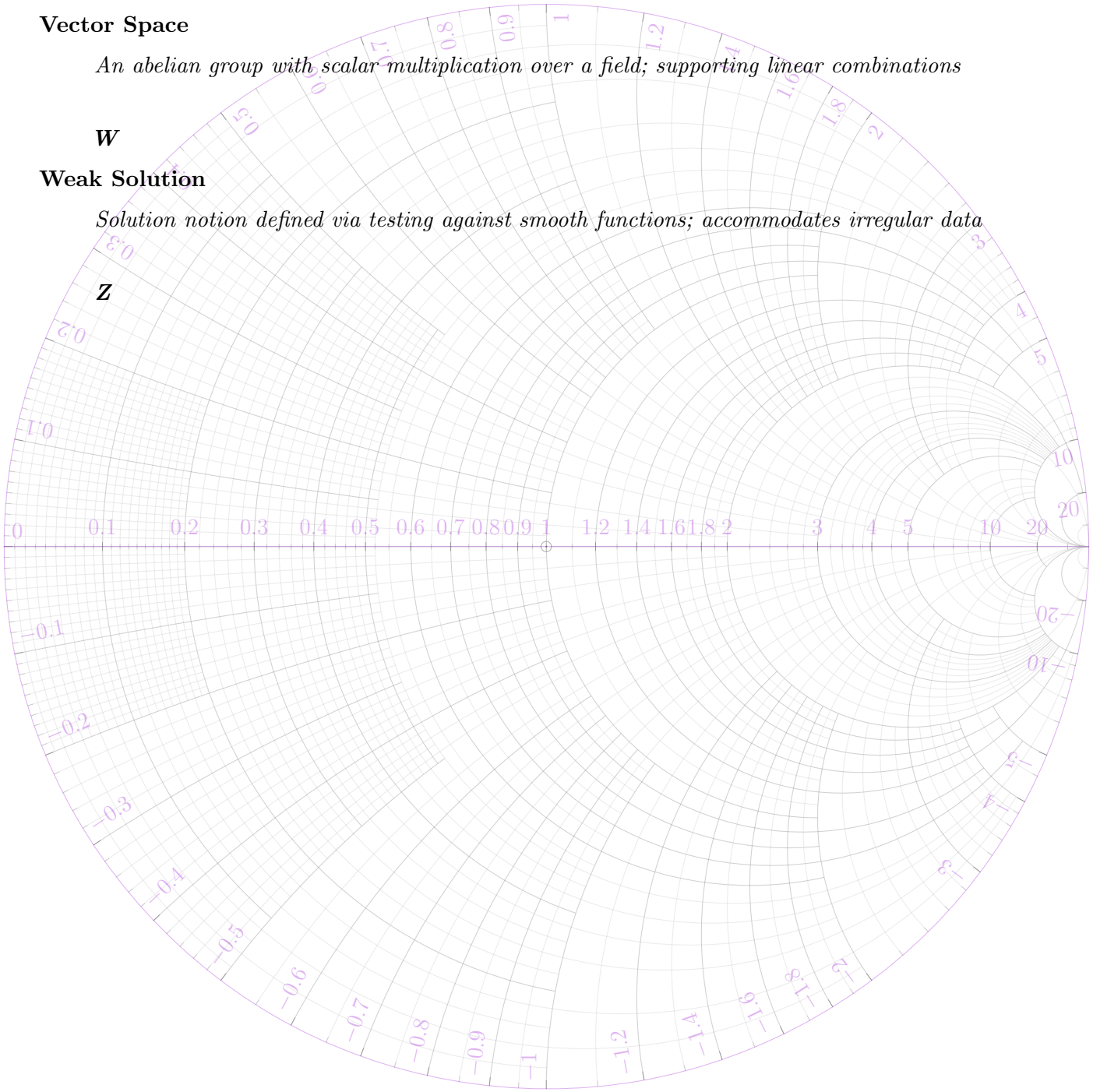
An abelian group with scalar multiplication over a field; supporting linear combinations

W

Weak Solution

Solution notion defined via testing against smooth functions; accommodates irregular data

Z



Software and Hardware

A

Access Control

Measures to ensure that only authorised users are allowed to use information systems, services, and resources (ISO 27001)

Asset

Anything that has value to the organisation and must therefore be protected, including information, software, hardware, services, and people (ISO 27001)

Audit

A systematic, independent, documented process for obtaining objective evidence and evaluating it against criteria (ISO 9001, ISO 27001)

C

Certification

Third-party attestation that an organisations management system conforms to requirements of a standard (ISO 9001, ISO 27001)

Competence

Ability to apply knowledge and skills to achieve intended results in processes and products (ISO 9001)

Continual Improvement

Recurring activity to enhance performance and management system effectiveness (ISO 9001, ISO 27001)

Corrective Action

Action to eliminate the cause of nonconformity and prevent recurrence (ISO 9001)

I

Information Security

Preservation of confidentiality, integrity, and availability of information (ISO 27001)

Integrity

Accuracy and completeness of information and assets (ISO 27001)

Interested Party

A person or organisation that can affect, be affected by, or perceive itself to be affected by a decision or activity (ISO 9001, ISO 27001)

M

Management System

A set of interrelated or interacting elements of an organisation to establish policies, objectives, and processes (ISO 9001, ISO 27001)

Monitoring

Determining the status of a system, process, or activity (ISO 9001, ISO 27001)

N

Nonconformity

Non-fulfilment of a requirement (ISO 9001, ISO 27001)

O

Objective Evidence

Data supporting the existence or verity of something (ISO 9001)

P

Performance Evaluation

Determination of the effectiveness of a management system through monitoring, measurement, analysis, and evaluation (ISO 9001)

Policy

Intentions and direction of an organisation as formally expressed by top management (ISO 9001, ISO 27001)

Preventive Action

Action taken to eliminate the cause of a potential nonconformity (ISO 9001)

Process

A set of interrelated or interacting activities that transforms inputs into outputs (ISO 9001)

R

Requirement

A need or expectation that is stated, generally implied, or obligatory (ISO 9001, ISO 27001)

S

Scope

Boundaries and applicability of the management system (ISO 9001, ISO 27001)

Statement of Applicability (SoA)

Documented statement describing control objectives and controls relevant to the organisation (ISO 27001)

Supplier

Organisation or person that provides a product or service (ISO 9001)

T

Top Management

Person or group of people who direct and control an organisation at the highest level (ISO 9001, ISO 27001)

Traceability

Ability to trace the history, application, or location of an item through recorded identification (ISO 9001)

V

Validation

Confirmation that requirements for a specific intended use or application have been fulfilled (ISO 9001)

Verification

Confirmation that specified requirements have been fulfilled (ISO 9001)

Z

Risks

R

Risk Action Request

Recommended treatment alternatives and supporting information for one or more risks determined to be above a risk threshold [ISO/IEC 16085:2006]

Risk Analysis

Process of examining identified risk factors for probability of occurrence, potential loss, and potential risk-handling strategies

Risk Appetite

The degree of uncertainty an entity is willing to take on, in anticipation of a reward [PMBOK Guide, Fifth Edition]

Risk Audit

Examination and documentation of the effectiveness of risk responses in dealing with identified risks and their root causes, as well as the effectiveness of the risk management process [PMBOK Guide, Fifth Edition]

Risk Avoidance

Course of action that removes a risk factor from further consideration. A risk response strategy whereby the project team acts to eliminate the threat or protect the project from its impact [PMBOK Guide, Fifth Edition]

Risk Breakdown Structure (RBS)

A hierarchical representation of risks according to their risk categories [PMBOK Guide, Fifth Edition]

Risk Categorisation

Organisation by sources of risk (e.g., using the RBS), the area of the project affected (e.g., using the WBS), or other useful category (e.g., project phase) to determine the areas of the project most exposed to the effects of uncertainty [PMBOK Guide, Fifth Edition]

Risk Category

A group of potential causes of risk [PMBOK Guide, Fifth Edition]

Risk Criteria

Terms of reference by which the significance of risk is assessed [ISO/IEC 16085:2006]. Terms of reference against which the significance of a risk is evaluated [ISO/IEC 15026-3:2015]

Risk Data Quality Assessment

Technique to evaluate the degree to which the data about risks is useful for risk management [PMBOK Guide, Fifth Edition]

Risk Exposure

Potential loss presented to an individual, project, or organisation by a risk [ISO/IEC 16085:2006]. Function of the likelihood that the risk will occur and the magnitude of the consequences of its occurrence [ISO/IEC 16085:2006]. Product of probability times potential loss for a risk factor

Risk Factor

Potential problem that would be detrimental to a planned activity, project, or program, characterised by the probability of problem occurrence and a potential loss if the problem occurs

Risk Handling

Course of action taken in response to a risk factor, including acceptance, avoidance, transfer, and mitigation

Risk Identification

Organised, systematic approach to determining the risk factors associated with a planned activity, project, or program

Risk Leverage Factor (RLF)

RLF = (REB REA)/RMC, where REB is risk exposure before mitigation, REA is risk exposure after mitigation, and RMC is the mitigation cost

Risk Management

Organised process for identifying and handling risk factors. Analytic process to identify what might cause harm or loss; assess and quantify risks; and develop or implement approaches to prevent or handle causes of risk [ISO/IEC 16085:2006; PMBOK Guide, Fifth Edition]

Risk Management Plan

Description of how the elements and resources of the risk management process will be implemented within an organisation or project [ISO/IEC 16085:2006]. Component of the management plan that describes how risk management activities will be structured and performed [PMBOK Guide, Fifth Edition]

Risk Management Process

Continuous process for systematically identifying, analysing, treating, and monitoring risk throughout the life cycle of a product or service [ISO/IEC 16085:2006]

Risk Management System

Set of elements of an organisations management system concerned with managing risk [ISO/IEC 16085:2006]

Risk Metric

Objective measure associated with a risk factor to be mitigated

Risk Mitigation

A risk response strategy whereby the project team acts to reduce the probability of occurrence or impact of a risk [PMBOK Guide, Fifth Edition]

Risk Monitoring and Control

Tracking identified risks, monitoring residual risks, identifying new risks, executing response plans, and evaluating effectiveness throughout the project life cycle

Risk Profile

Chronological record of a risks current and historical state information [ISO/IEC 16085:2006]

Risk Reassessment

Identification of new risks, reassessment of current risks, and the closing of outdated risks [PMBOK Guide, Fifth Edition]

Risk Reduction

Reducing the probability and/or potential impact of a risk factor

Risk Reduction Measure

Steps taken to reduce or mitigate risk [ISO/IEC 15026-3:2015]

Risk Register

Document in which the results of risk analysis and risk response planning are recorded [PMBOK Guide, Fifth Edition]

Risk Source

Element that, alone or in combination, has the intrinsic potential to give rise to risk [ISO/IEC 15026-3:2015]

Risk State

Current project risk information relating to an individual risk [ISO/IEC 16085:2006]

Risk Threshold

Measure of the level of uncertainty or impact at which a stakeholder may have a specific interest [PMBOK Guide, Fifth Edition; ISO/IEC 16085:2006]

Risk Tolerance

The degree, amount, or volume of risk that an organisation or individual will withstand [PMBOK Guide, Fifth Edition]

Risk Transfer

Transferring responsibility for managing a risk factor to another organisation or functional entity better able to mitigate it

Risk Transference

Risk response strategy whereby the project team shifts the impact of a threat to a third party, together with ownership of the response [PMBOK Guide, Fifth Edition]

Risk Treatment

Process of selection and implementation of measures to modify risk [ISO/IEC 16085:2006]. Means of modifying risk. Process to eliminate risk or reduce it to a tolerable level [ISO/IEC 15026-3:2015]

Risk Trigger

Predetermined threshold value of a risk metric that triggers invocation of a contingency plan when the metric crosses the threshold

Risk Urgency Assessment

Review and determination of the timing of actions which may need to occur sooner than other risk items [PMBOK Guide, Fifth Edition]

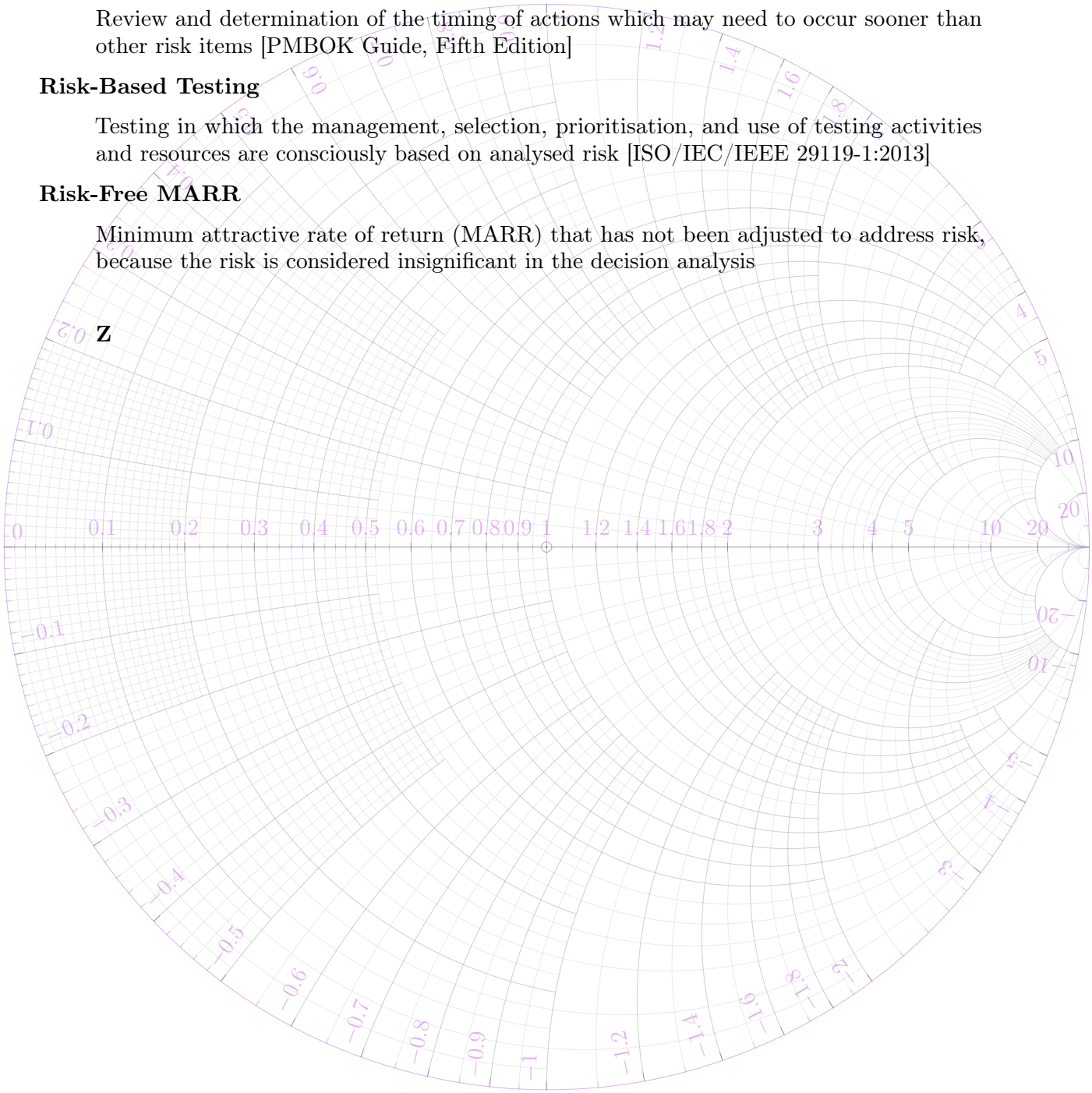
Risk-Based Testing

Testing in which the management, selection, prioritisation, and use of testing activities and resources are consciously based on analysed risk [ISO/IEC/IEEE 29119-1:2013]

Risk-Free MARR

Minimum attractive rate of return (MARR) that has not been adjusted to address risk, because the risk is considered insignificant in the decision analysis

Z



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